# Sequential Monte Carlo$^2$ for Bayesian Filtering and Stochastic Optimal Control of a Physiological SDE

Lecture notes on the FSA-v2 model

Ajay Talati

Generated 29 April 2026

Source: `python-smc2-filtering-control` at commit `684dd49`

(github.com/ajaytalati/python-smc2-filtering-control)

**Abstract**

These notes present the mathematical theory underlying a research code base that combines particle filtering and stochastic optimal control on a three-dimensional physiological stochastic differential equation. We define the FSA-v2 model in its G1-reparametrized form, prove that all 30 static parameters are locally identifiable from a one-day window of four-channel observations by analytical Fisher-information-matrix calculation, and establish exponential ergodicity of the SDE on its operating domain via a quadratic Lyapunov function. We then develop the bootstrap particle filter and the SMC$^2$ algorithm of Chopin, Jacob and Papaspiliopoulos, recast optimal scheduling of training stimulus as posterior inference under a tempered cost likelihood, and close the loop in a model-predictive controller that interleaves a rolling SMC$^2$ filter with a posterior-mean control SMC$^2$ over an RBF schedule. The receding-horizon section is presented in both abstract mathematical and code-level form, and includes a linear–quadratic–Gaussian / differential-Riccati alternative that is sketched in full but flagged as the principal piece of remaining analytical work.

The audience is graduate students in applied mathematics, mathematical engineering, applied stochastics, or process engineering. Familiarity with Itô stochastic differential equations, Bayesian filtering, and basic optimal control at the level of a one-semester course is assumed; the particle-filtering and tempered-SMC machinery is built up from first principles.

## Contents

**1 Introduction** **3**

**2 The FSA-v2 SDE model** **4**
- 2.1 State, drift, and diffusion . . . 4
- 2.2 Truth parameters . . . 5
- 2.3 The training-stimulus envelope . . . 5
- 2.4 Four-channel observation model . . . 6

**3 Identifiability via the Fisher information matrix** **6**
- 3.1 Fisher information and local identifiability . . . 6
- 3.2 The pre-G1 model has three rank-deficient pairs . . . 7
- 3.3 The G1 reparametrization absorbs the three pairs . . . 7
- 3.4 Empirical evidence of identifiability after G1 . . . 7

**4 Stability of the FSA-v2 SDE: a Lyapunov analysis** **8**
- 4.1 Deterministic skeleton: equilibria and the Stuart–Landau bifurcation . . . 8
- 4.2 Stochastic stability and exponential ergodicity . . . 9

**5 Numerical stability of the SDE solver** **10**
5.1 The solver . . . 10
5.2 What "numerical stability" means here . . . 10
5.3 Linearising the FSA-v2 SDE . . . 11
5.4 Stability margin at the current step size . . . 11
5.5 What does bind, then? . . . 11
5.6 Recommendation . . . 12

**6 The bootstrap particle filter** **12**
6.1 Bayesian filtering recursion . . . 12
6.2 Sequential importance resampling . . . 12
6.3 The unbiased marginal likelihood . . . 13

**7 SMC$^2$ over parameters** **13**
7.1 The tempered posterior over $\theta_{\text{dyn}}$ . . . 13
7.2 Adaptive tempering and the move step . . . 14
7.3 Rolling-window SMC$^2$ and the Schrödinger–Föllmer bridge . . . 14

**8 Control as inference** **15**
8.1 The cost functional . . . 15
8.2 Schedule decoder . . . 15
8.3 Tempered control posterior . . . 15
8.4 Posterior mean as the controller output . . . 16

**9 Receding-horizon model-predictive control** **16**
9.1 Finite-horizon stochastic optimal control . . . 16
9.2 The receding-horizon principle . . . 16
9.3 Stochastic and output-feedback variants . . . 17
9.4 The FSA-v2 MPC pipeline — mathematical statement . . . 17
9.5 Code-level structure of the MPC implementation . . . 18
9.6 Recursive feasibility and the F-barrier . . . 19
9.7 Robustness and the flat-cost-surface argument . . . 19
9.8 An exact-approximate alternative: LQG and the differential Riccati equation . . . 20

**10 Results** **21**
10.1 Stage D: open-loop schedule on the 84-day horizon . . . 21
10.2 Stage E5: closed-loop MPC on the 14-day macrocycle . . . 21

**11 Discussion and open questions** **22**
11.1 Resolved: Fisher information rank identifiability . . . 22
11.2 Resolved: Lyapunov stability of the SDE . . . 22
11.3 Outstanding: LQG / Riccati exact-approximate alternative . . . 23
11.4 Other open questions . . . 23

**A Fisher information rank deficiencies of the pre-G1 model: derivations** **24**
A.1 Pair 1: $(\kappa_B, \varepsilon_A)$ . . . 24
A.2 Pair 2: $(\mu_F, \mu_{FF})$ . . . 25
A.3 Pair 3: $(\tau_F, \lambda_A)$ . . . 25
A.4 The null space of the FIM . . . 25
A.5 The G1 reparametrization is mathematically equivalent to the pre-G1 model . . . 25

**B Stability of the FSA-v2 SDE: derivations** **26**
B.1 The drift Jacobian at the $A = 0$ branch . . . 26
B.2 The bifurcation locus . . . 27
B.3 Foster–Lyapunov drift inequality (proof of Theorem 7) . . . 27

**C Numerical stability of the SDE solver: derivations** **29**
C.1 The linear test SDE and explicit Euler–Maruyama . . . 29
C.2 Component-by-component bounds for FSA-v2 . . . 30
C.3 The role of the inner substepping . . . 30
C.4 The CIR diffusion and the Feller condition . . . 30
C.5 What an enlarged step would buy . . . 31

# 1 Introduction

A human athlete is, from the point of view of an applied mathematician, a non-linear stochastic dynamical system whose state is only ever observed through a handful of noisy proxies. Heart rate, sleep duration, perceived stress, and step counts are all loosely informative about the latent fitness, fatigue, and autonomic condition of the body, but none of them directly measures any of these quantities. The classical Banister model of training response (Banister, 1991) treats the athlete as a deterministic two-compartment system; the present notes work with a stochastic three-compartment refinement, denoted FSA-v2, in which fitness, fatigue, and an autonomic amplitude evolve as a coupled Itô stochastic differential equation driven by a single exogenous control — the daily training stimulus $\Phi(t)$.

Two questions follow naturally:

1. *Filtering.* Given a stream of multi-channel daily observations, what is the joint posterior distribution over the latent state $(B, F, A)$ and the static parameters $\theta$ of the SDE?

2. *Control.* Given the same posterior, what schedule $\Phi : [0, T] \to [0, \Phi_{\max}]$ maximises a chosen functional of the latent trajectory, subject to a soft constraint that fatigue $F(t)$ should not exceed a physiological ceiling $F_{\max}$?

Both questions sit at the intersection of Bayesian inference and stochastic optimal control. They are usually treated as separate disciplines; one of the contributions of the underlying code base is to express both within a single sequential Monte Carlo framework. The filter is built from a bootstrap particle filter, lifted to a sampler over static parameters by the SMC$^2$ algorithm (Chopin, Jacob and Papaspiliopoulos, 2013) and operated in a rolling-window mode. The controller is built by recasting the cost-minimisation problem as posterior inference under a tempered cost likelihood, in the spirit of Toussaint (2009) and the more general control-as-inference literature (Kappen, 2005; Levine, 2018). The two machines are then composed in a model-predictive controller that re-plans the next twelve hours of training stimulus from the most recent posterior.

These notes are organised so that the mathematical content can be read linearly. Section 2 defines the FSA-v2 SDE and observation model in their G1-reparametrized form. Section 3 carries out the Fisher-information-matrix analysis: an analytical derivation of the three rank-deficient parameter pairs of the original parametrization, the reparametrization that absorbs them, and the empirical evidence that the resulting model is fully identifiable at one-day windows. Section 4 establishes stability of the SDE: Jacobian-based local analysis of the deterministic skeleton, the Stuart–Landau bifurcation locus, and a quadratic Lyapunov function satisfying a Foster–Lyapunov drift inequality, from which exponential ergodicity follows. Section 5 bridges from continuous-time theory to numerical practice: it asks whether the fifteen-minute outer step of the explicit Euler–Maruyama solver actually used in the code base is small enough for

mean-square stability, finds the answer to be "yes, with three orders of magnitude of safety margin", and identifies the resolution of the sub-daily $\Phi$-burst as the binding constraint should one wish to enlarge the step to speed up long-horizon experiments. Section 6 develops the bootstrap particle filter. The $\mathrm{SMC}^2$ outer loop and its rolling-window variant are presented in Section 7. Section 8 formulates the control problem as a posterior-inference problem. Section 9 is the extended core of the document: it develops the receding-horizon principle, describes the FSA-v2 model-predictive controller in both mathematical and implementation terms, and sketches a linear–quadratic–Gaussian approximation that could be added as a cheap baseline. Section 10 reports the two headline empirical results of the code base: an open-loop gain of +28% in mean autonomic amplitude over a constant-stimulus baseline on an 84-day horizon, and a closed-loop gain of +17% in a fourteen-day rolling controller. Section 11 closes by reviewing the identifiability and stability analyses just completed, and by identifying the LQG / Riccati alternative as the principal piece of remaining theoretical work.

# 2 The FSA-v2 SDE model

## 2.1 State, drift, and diffusion

Let $(B(t),\, F(t),\, A(t))$ denote the latent state of a single individual at time $t \geq 0$, measured in days. The three components are interpreted as

- $B(t) \in [0, 1]$ – a normalised *fitness* variable; the Banister chronic compartment;
- $F(t) \in [0, \infty)$ – an unnormalised *fatigue* variable; the Banister acute compartment;
- $A(t) \in [0, \infty)$ – the *amplitude* of an autonomic / circadian Stuart–Landau oscillator.

The single exogenous control input is the (non-negative) training-stimulus rate $\Phi(t) \in [0, \Phi_{\max}]$, expressed in arbitrary daily TRIMP-like units.

The model is presented in the *G1-reparametrized* form. As we will see in Section 3, an earlier form of this model contained three pairs of parameters that the four-channel observation model could not separate. The G1 form rotates each pair into one strongly identified parameter and one residual; the drift formulas below are mathematically equivalent to the older form at a chosen operating point $(B_{\mathrm{typ}}, F_{\mathrm{typ}}, A_{\mathrm{typ}})$ that we now fix.

**Definition 1** (Operating point)**.** We work throughout at the reference operating point

$$A_{\mathrm{typ}} = 0.10, \qquad F_{\mathrm{typ}} = 0.20, \qquad \Phi_{\mathrm{typ}} = 1.0. \tag{1}$$

This is the regime of a de-trained athlete training at the canonical Banister default load. The reparametrization is centred at this point, and prior beliefs on the residual parameters encode "the system is typically near here."

The drift coefficients are

$$\mu(B, F) = \mu_0 + \mu_B B - \mu_F F - \mu_{FF}\,(F - F_{\mathrm{typ}})^2, \tag{2}$$

$$\mathrm{d}B = \left[\, \kappa_B \cdot \frac{1 + \varepsilon_A A}{1 + \varepsilon_A A_{\mathrm{typ}}}\, \Phi(t) - \frac{B}{\tau_B} \,\right] \mathrm{d}t + \sigma_B \sqrt{B(1-B)}\, \mathrm{d}W_B, \tag{3}$$

$$\mathrm{d}F = \left[\, \kappa_F\, \Phi(t) - \frac{1 + \lambda_A A}{1 + \lambda_A A_{\mathrm{typ}}}\, \frac{F}{\tau_F} \,\right] \mathrm{d}t + \sigma_F \sqrt{F}\, \mathrm{d}W_F, \tag{4}$$

$$\mathrm{d}A = \left[\, \mu(B, F)\, A - \eta\, A^3 \,\right] \mathrm{d}t + \sigma_A \sqrt{A}\, \mathrm{d}W_A, \tag{5}$$

where $W_B, W_F, W_A$ are independent scalar Brownian motions.

The diffusions in (3)–(4) are unchanged from the pre-G1 model: a Jacobi diffusion on $[0, 1]$ for $B$ (Forman and Sørensen, 2008) whose volatility vanishes at both endpoints, and square-root (CIR-type, Cox–Ingersoll–Ross 1985) diffusions for $F$ and $A$ whose volatility vanishes at the lower boundary.

**Reading the residual factors.** The two ratios $(1 + \varepsilon_A A)/(1 + \varepsilon_A A_{\text{typ}})$ and $(1 + \lambda_A A)/(1 + \lambda_A A_{\text{typ}})$ each equal one when $A = A_{\text{typ}}$. So at the operating point, and for the constant control $\Phi \equiv \Phi_{\text{typ}}$, the drift collapses to

$$\dot{B} = \kappa_B \Phi - B/\tau_B, \qquad \dot{F} = \kappa_F \Phi - F/\tau_F, \qquad \mu = \mu_0 + \mu_B B - \mu_F F. \tag{6}$$

That is, the strongly identified parameters $\kappa_B, \tau_F, \mu_F$ play exactly the role they would in a Banister model linearised at $(F_{\text{typ}}, A_{\text{typ}})$. The residuals $\varepsilon_A, \lambda_A, \mu_{FF}$ control departures from the operating point, and they do so via terms that vanish at the operating point itself; this is what makes them weakly identified at one-day windows and consequently candidates for tighter priors.

## 2.2 Truth parameters

Table 1 lists the truth parameters of the G1-reparametrized model. Compared to the pre-G1 form, three values change: $\kappa_B = 0.012 \cdot (1 + \varepsilon_A A_{\text{typ}}) = 0.01248$, $\tau_F = 7.0/(1 + \lambda_A A_{\text{typ}}) = 6.\overline{36}\,\text{d}$, and $\mu_F = 0.10 + 2F_{\text{typ}}\mu_{FF} = 0.26$, with a small additive shift $\mu_0 \mapsto \mu_0 + \mu_{FF} F_{\text{typ}}^2 = 0.036$ absorbing the constant from completing the square in (2).

| Group | Symbol | Value | Meaning |
|---|---|---|---|
| Banister timescales | $\tau_B$ | 42.000 | chronic time constant (d) |
| | $\tau_F$ | 6.364 | effective acute time constant (d) |
| Banister gains | $\kappa_B$ | 0.01248 | effective $B$-gain at $A_{\text{typ}}$ |
| | $\kappa_F$ | 0.030 | $F$-gain per unit $\Phi$ |
| $A$-coupling (residual) | $\varepsilon_A$ | 0.40 | residual $B$-gain boost beyond $A_{\text{typ}}$ |
| | $\lambda_A$ | 1.00 | residual $F$-clearance boost beyond $A_{\text{typ}}$ |
| Stuart–Landau | $\mu_0$ | 0.036 | growth-rate intercept |
| | $\mu_B$ | 0.30 | fitness raises $\mu$ |
| | $\mu_F$ | 0.26 | local slope of $\mu$ at $F_{\text{typ}}$ |
| | $\mu_{FF}$ | 0.40 | residual curvature of $\mu$ in $F$ |
| | $\eta$ | 0.20 | cubic damping |
| Diffusion | $\sigma_B$ | 0.010 | Jacobi noise scale |
| | $\sigma_F$ | 0.012 | CIR noise scale on $F$ |
| | $\sigma_A$ | 0.020 | CIR noise scale on $A$ |

Table 1: Truth parameters of the G1-reparametrized FSA-v2 dynamics. Source: `version_2/models/fsa_high_res/_dynamics.py`.

## 2.3 The training-stimulus envelope

The control variable in the SDE is the instantaneous training-stimulus rate $\Phi(t)$. In practice training is delivered in discrete daily sessions concentrated in the morning, and the simulator expands a daily schedule $(\Phi_d)_{d=1,\ldots,N}$ into a high-resolution envelope at fifteen-minute resolution as follows. Let $\tau_{\text{burst}} = 3\,\text{h}$ be a shape parameter, and let $[t_{\text{wake}}, t_{\text{sleep}}] = [7, 23]$ be the daily wake window (in hours). The within-day envelope is

$$e(t) \;\propto\; \begin{cases} (t - t_{\text{wake}})\,\exp(-(t - t_{\text{wake}})/\tau_{\text{burst}}) & t_{\text{wake}} \le t < t_{\text{sleep}}, \\ 0 & \text{otherwise}, \end{cases} \tag{7}$$

which is the standard Gamma($k = 2$) shape, peaking at $t - t_{\text{wake}} = \tau_{\text{burst}}$. The constant of proportionality is fixed so that $\int_{\text{day}} \Phi(t)\,\mathrm{d}t = 24\,\Phi_d$, i.e. the daily integral of the within-day envelope is preserved. The slow Banister equations (3)–(4) therefore see the same daily load whether $\Phi$ is delivered as a constant or as a morning-loaded burst, and the fast Stuart–Landau dynamics in $A$ pick up the within-day shape.

### 2.4 Four-channel observation model

The state $(B, F, A)$ is not directly observed. Instead we receive four discrete-time channels at the fifteen-minute grid:

$$\mathrm{HR}(t) \sim \mathcal{N}\big(\,\mathrm{HR}_{\text{base}} - \kappa_B^{\text{HR}}\,B + \alpha_A^{\text{HR}}\,A + \beta_C^{\text{HR}}\,C(t),\ \sigma_{\text{HR}}^2\big), \qquad \text{(sleep-gated)} \tag{8}$$

$$P(\text{sleep}(t) = 1 \mid A, C) = \sigma\big(\,k_C\,C(t) + k_A\,A - \tilde{c}\,\big), \qquad \text{(always observed)} \tag{9}$$

$$S(t) \sim \mathcal{N}\big(\,S_{\text{base}} + k_F\,F - k_A^S\,A + \beta_C^S\,C(t),\ \sigma_S^2\big), \qquad \text{(wake-gated)} \tag{10}$$

$$\log(\text{steps}(t) + 1) \sim \mathcal{N}\big(\,\mu_0^{\text{step}} + \beta_B^{\text{step}}\,B - \beta_F^{\text{step}}\,F + \beta_A^{\text{step}}\,A + \beta_C^{\text{step}}\,C(t),\ \sigma_{\text{step}}^2\big), \qquad \text{(wake-gated)} \tag{11}$$

where $C(t) = \cos(2\pi t + \phi)$ is a fixed deterministic circadian covariate (not a latent variable). "Sleep-gated" channels are emitted only when the sleep label of (9) is one; "wake-gated" channels only when it is zero. The collection of observation-model parameters $(\mathrm{HR}_{\text{base}}, \kappa_B^{\text{HR}}, \ldots, \sigma_{\text{step}})$ brings the full static parameter vector $\theta_{\text{dyn}}$ for the joint state-space model to 30 scalar entries. We write $y_{1:n}$ for the multi-channel observation record up to time $t_n$ and $p(y_{1:n} \mid \theta_{\text{dyn}})$ for its marginal likelihood under the model.

The two analyses that an earlier version of these notes flagged as outstanding – Fisher Information rank identifiability and Lyapunov stability – have since been carried out and form Sections 3 and 4 respectively.

## 3 Identifiability via the Fisher information matrix

A model is no use if its parameters cannot be recovered from the observations: an inference procedure may converge confidently to one of an infinite family of indistinguishable parameter vectors. This section asks whether the FSA-v2 model defined in Section 2 suffers from such a pathology, settles the matter analytically, and reports the empirical fingerprint of the diagnosis. The detailed derivations of the rank-deficient parameter pairs and of the G1 reparametrization are deferred to Appendix A.

### 3.1 Fisher information and local identifiability

For a parameter vector $\theta \in \mathbb{R}^d$ and an observation record $y$, the score function is $s(\theta; y) = \nabla_\theta \log p(y \mid \theta)$ and the *Fisher information matrix* (FIM) is

$$\mathcal{I}(\theta) \;=\; \mathbb{E}_{y \sim p(\cdot \mid \theta)}\big[\,s(\theta; y)\,s(\theta; y)^\top\big]. \tag{12}$$

$\mathcal{I}(\theta)$ is symmetric positive semidefinite, and two facts about it explain why it is the right diagnostic.

**Proposition 2** (Local identifiability)**.** *Suppose $\theta \mapsto p(y \mid \theta)$ is twice differentiable in a neighbourhood of $\theta^*$. Then $\theta^*$ is locally identifiable if and only if $\mathcal{I}(\theta^*)$ has full rank $d$.*

**Proposition 3** (Cramér–Rao)**.** *Any unbiased estimator $\hat{\theta}$ of $\theta$ has covariance bounded below in the positive-semidefinite ordering by $\mathcal{I}(\theta)^{-1}$ (when $\mathcal{I}$ is invertible).*

In words: the FIM *rank* tells you whether the data can pin $\theta$ down at all, and the FIM *inverse* tells you how sharply. A rank-deficient FIM means there is a direction in parameter space along which the data are silent – two distinct $\theta$ vectors produce indistinguishable observation distributions – and any procedure trying to estimate $\theta$ in that direction will be at the mercy of the prior, not the data.

### 3.2 The pre-G1 model has three rank-deficient pairs

The four-channel observation model (8)–(11) probes the latent state $(B, F, A)$ through linear-in-state likelihoods, and the latent state in turn carries information about the dynamics parameters via the drift (3)–(5). At a one-day window the slow Banister state moves only marginally, so the inferential signal about the dynamics parameters is concentrated in how $\partial(B, F, A)/\partial\theta$ couples through the observation likelihood.

A direct calculation along these lines (deferred to Appendix A) reveals that, near the operating point of Definition 1, three pairs of pre-G1 parameters enter the one-day likelihood through a single linear combination each, so that the contribution of each pair to the FIM is rank-one rather than rank-two. The three pairs and their identifiable combinations are summarised in Table 2.

The intuitive story behind each pair is the same: at one-day windows the system rarely strays far from the operating point, so any parameter whose effect manifests only via departures from that operating point is invisible to the data. For instance, in the pair $(\kappa_B, \varepsilon_A)$ the gain $\kappa_B$ multiplies the daily training stimulus, while $\varepsilon_A$ multiplies $\kappa_B$ by a factor close to one whenever $A \approx A_{\text{typ}}$. Only the *product* $\kappa_B(1 + \varepsilon_A A_{\text{typ}})$ is felt by the data; the individual factors are not.

### 3.3 The G1 reparametrization absorbs the three pairs

The cure is a simple coordinate change: rotate each rank-deficient pair into a strongly identified linear combination and a residual that vanishes at the operating point. The G1 reparametrization, already shown in equations (2)–(5), performs this rotation. Table 2 lists the three combinations side by side.

| Pair | Strongly identified | Residual | Vanishing condition |
|---|---|---|---|
| $(\kappa_B, \varepsilon_A)$ | $\kappa_B^{\text{new}} = \kappa_B(1 + \varepsilon_A A_{\text{typ}})$ | $\varepsilon_A$ | $A = A_{\text{typ}}$ |
| $(\mu_F, \mu_{FF})$ | $\mu_F^{\text{new}} = \mu_F + 2F_{\text{typ}}\mu_{FF}$ | $\mu_{FF}$ | $F = F_{\text{typ}}$ |
| $(\tau_F, \lambda_A)$ | $\tau_F^{\text{new}} = \tau_F/(1 + \lambda_A A_{\text{typ}})$ | $\lambda_A$ | $A = A_{\text{typ}}$ |

Table 2: The Stage G1 reparametrization. Each rank-deficient parameter pair of the pre-G1 model is rotated into a strongly identified combination (left column) and a residual (middle column) that vanishes at the operating point of Definition 1. The parameter *names* are kept; only the meanings and truth values change. There is also an implicit shift $\mu_0 \mapsto \mu_0 + \mu_{FF}F_{\text{typ}}^2$ that absorbs the constant from completing the square in (2).

The reparametrized drift is mathematically equivalent to the pre-G1 drift under a fixed relabelling of parameters; the equivalence (stated as Proposition 16 in Appendix A) is verified numerically by a unit test in `tests/test_g1_reparam.py` that confirms the two drift formulas agree to floating-point precision over 50 random states.

### 3.4 Empirical evidence of identifiability after G1

The analytical argument predicts full FIM rank at the operating point. The implementation furnishes empirical confirmation at one-day rolling windows on simulated truth data.

| Quantity | Pre-G1 | Post-G1 |
|---|---|---|
| rank$\widehat{\mathcal{I}}$ at one-day window | $d-3$ | $d$ |
| parameters covered by 90 % credible interval | 27/30 | **30/30** |
| identifiable estimable parameters | 3/6 | **6/6** |
| posterior std on $\varepsilon_A$ (lognormal scale) | 0.093 | **0.016** |
| 90 % CI for $\varepsilon_A$ | [0.275, 0.550] | [**0.373**, **0.423**] |

Table 3: Identifiability before and after the G1 reparametrization, measured at one-day rolling-window scale on simulated truth data. "Estimable" counts the six parameters that the FIM analysis identified as candidates for tightening. Numbers from `version_2/outputs/fsa_high_res/` and the Stage G1 commit record. The residual standard deviation on $\varepsilon_A$ tightens by roughly a factor of six; analogous tightening is observed on the other two residuals $\mu_{FF}$ and $\lambda_A$.

The numbers in Table 3 confirm the analytical prediction. The G1 form is identifiable at one-day windows; the strongly identified parameters $(\kappa_B, \tau_F, \mu_F)$ retain their original prior widths; and the residuals $(\varepsilon_A, \lambda_A, \mu_{FF})$, whose posteriors were formerly artificially wide because of the rank deficiency, now contract to roughly a sixth of their pre-G1 spread under tightened priors that the rank analysis itself motivates.

*Remark* 4 (Where the FIM analysis lives in the code). The analytical FIM derivation summarised here – and developed in full in Appendix A – is recorded in the project's commit log (Stage G1) rather than as standalone code. A useful follow-up would be to add a numerical FIM module under `smc2fc/diagnostics/fim/` that reproduces these results from the JAX dynamics by automatic differentiation, so that any future modification of the model triggers a re-run of the rank check automatically.

# 4 Stability of the FSA-v2 SDE: a Lyapunov analysis

Identifiability tells us that the inference problem for $\theta$ is well-posed; stability tells us that the dynamical system itself is well-behaved. This section asks two stability questions about the FSA-v2 SDE in its G1-reparametrized form: where are the equilibria of its deterministic skeleton, and is the full stochastic process ergodic on its physiological operating domain? The first question admits a clean answer in terms of the Stuart–Landau growth rate; the second is settled affirmatively via a quadratic Lyapunov function. Detailed Jacobian computations, the Foster–Lyapunov drift calculation, and bounds on constants are deferred to Appendix B.

## 4.1 Deterministic skeleton: equilibria and the Stuart–Landau bifurcation

Set $\sigma_B = \sigma_F = \sigma_A = 0$ in (3)–(5) and a constant control $\Phi(t) \equiv \Phi_c$. The deterministic skeleton has an explicit $A = 0$ branch

$$B_0^*(\Phi_c) = \frac{\kappa_B \tau_B \Phi_c}{1 + \varepsilon_A A_{\text{typ}}}, \quad F_0^*(\Phi_c) = \kappa_F \tau_F (1 + \lambda_A A_{\text{typ}})\, \Phi_c, \quad A_0^* = 0, \tag{13}$$

together with an implicit oscillatory branch where $A^2 = \mu(B, F)/\eta$. Local stability of the $A = 0$ branch is determined entirely by the sign of the Stuart–Landau growth rate at the equilibrium, as the following result records.

**Proposition 5** (Local stability of the $A = 0$ branch)**.** *The deterministic equilibrium* (13) *is locally asymptotically stable if and only if* $\mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) < 0$.

The proof is short: the drift Jacobian at (13) turns out to be upper triangular, with two unconditionally negative eigenvalues coming from the slow Banister directions and a third eigenvalue equal to $\mu(B_0^*, F_0^*)$ controlling the autonomic amplitude. Details are in Appendix B.

The bifurcation locus $\{\Phi_c : \mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) = 0\}$ turns out to be a single positive root $\Phi_c^{\text{crit}} \approx 7.5$ TRIMP-units per day at the truth values of Table 1. The picture this paints is physiologically intuitive: at the canonical baseline $\Phi_c = 1$, $\mu(B_0^*, F_0^*) \approx 0.13 > 0$, so the $A = 0$ equilibrium is unstable and the model predicts that an athlete training at the default load settles on a non-trivial autonomic oscillation. Above the critical training load $\Phi_c \approx 7.5$, the quadratic overtraining-collapse term $\mu_{FF}(F - F_{\text{typ}})^2$ dominates, $\mu$ falls back through zero, and the system loses its oscillation: the autonomic amplitude collapses to zero, consistent with empirical observations that very large chronic training loads suppress autonomic variability.

### 4.2 Stochastic stability and exponential ergodicity

We turn to the full SDE (3)–(5) on the operating domain $\mathcal{X} = [0, 1] \times [0, \infty) \times [0, \infty)$. The state-dependent diffusions vanish at the boundary $\partial\mathcal{X}$, so $\mathcal{X}$ is positively invariant. We seek a single quantitative statement that captures the long-time behaviour of the process. The Foster–Lyapunov framework supplies the right object.

**Definition 6** (Foster–Lyapunov drift)**.** A function $V : \mathcal{X} \to [0, \infty)$ is said to satisfy a Foster–Lyapunov drift inequality with rate $\alpha > 0$ and constant $C \geq 0$ if the infinitesimal generator $\mathcal{L}$ of the SDE satisfies

$$\mathcal{L}V(x) \;\leq\; -\alpha\, V(x) + C \quad \text{for all } x \in \mathcal{X}. \tag{14}$$

Inequality (14) is the standard hypothesis that delivers exponential ergodicity in the sense of total-variation convergence to a unique invariant distribution (Khasminskii, 1980; Meyn and Tweedie, 1993). We exhibit a $V$ that satisfies it.

**Theorem 7** (Stochastic stability of FSA-v2)**.** *Under bounded controls $\Phi : \mathbb{R}_+ \to [0, \Phi_{\max}]$, the FSA-v2 SDE (3)–(5) satisfies the Foster–Lyapunov drift inequality (14) for the quadratic Lyapunov function*

$$V(B, F, A) \;=\; \tfrac{1}{2}(aB^2 + bF^2 + cA^2), \qquad a, b, c > 0, \tag{15}$$

*with rate $\alpha = \min\big(2/\tau_B,\, 1/[\tau_F(1 + \lambda_A A_{\text{typ}})]\big)$ and a finite constant $C$ depending on the truth parameters, $\Phi_{\max}$, and the choice of $a, b, c$.*

The proof is a routine but tedious computation of $\mathcal{L}V$ component by component, bounding each term using the boundedness of $B$ on $[0, 1]$ together with two applications of Young's inequality on the $F$ and $A$ components; we record it in Appendix B. The intuitive picture is that the three dissipative terms in the drift (linear decay of $B$, linear decay of $F$, cubic damping of $A$) overwhelm the linear growth and noise contributions outside a bounded region of state space, which is exactly what the Foster–Lyapunov inequality formalises.

**Theorem 8** (Khasminskii, 1980; Meyn–Tweedie, 1993)**.** *Let $X_t = (B_t, F_t, A_t)$ be the FSA-v2 process under a fixed bounded control $\Phi$, satisfying the conditions of Theorem 7. Then $X$ admits a unique invariant probability distribution $\pi^\Phi$ on $\mathcal{X}$, and there exist constants $K_0 > 0$ and $\rho \in (0, \alpha)$ such that, for every $x_0 \in \mathcal{X}$,*

$$\big\|\text{Law}(X_t \mid X_0 = x_0) - \pi^\Phi\big\|_{\text{TV}} \;\leq\; K_0\,(1 + V(x_0))\, e^{-\rho t} \quad \textit{for all } t \geq 0, \tag{16}$$

*where $\|\cdot\|_{\text{TV}}$ is the total-variation norm.*

The exponential rate $\rho$ inherits the slowest dissipative timescale of the model. With $\tau_B = 42$ days and $\tau_F^{\text{old}} := \tau_F(1 + \lambda_A A_{\text{typ}}) = 7$ days, the rate is $\alpha = 1/7\,\text{d}^{-1}$ and convergence to invariance has half-life roughly $7 \log 2 \approx 4.85$ days. This is consistent with the rolling-window choice of one-day windows and twelve-hour strides in Section 7: the SMC$^2$ filter operates on a fundamental dynamical scale comparable to the relaxation time of the process being filtered.

*Remark* 9 (Limitations of the bound). The proof of Theorem 7 in Appendix B uses the simplest possible quadratic Lyapunov function and crude global bounds on the residual factors of the drift. The constant $C$ that emerges is therefore loose: it scales with $\Phi^2_{\max}$ rather than with the time-average of $\Phi^2$, and ignores the cancellations that the real dynamics exhibit near the operating point. A sharper analysis – perhaps with a piecewise Lyapunov function that switches between the regimes $\mu < 0$ and $\mu > 0$ identified in Proposition 5 – would deliver a tighter rate and a smaller constant. Such a refinement is left for future work.

# 5 Numerical stability of the SDE solver

The continuous-time analyses of Sections 3 and 4 guarantee that the FSA-v2 model is well-posed and ergodic on its operating domain. Before passing those guarantees on to the inference machinery of the next four sections, we have to check that the *discretisation* of the SDE preserves them: a numerical solver with the wrong step size can introduce spurious blow-ups, suppress genuine fluctuations, or even reverse the sign of stability.

This section asks two practical questions. *Is the fifteen-minute time step currently used in the code base small enough to ensure numerical stability?* And, if so, *how much larger could it be made without losing stability?* – a question the reader will want to answer before running long-horizon experiments, where halving the step size doubles the wall-clock cost. The full algebra of the linear stability analysis, including a substepping sub-analysis for the cubic Stuart–Landau term and a discussion of the CIR diffusion's Feller boundary behaviour, is deferred to Appendix C.

## 5.1 The solver

The implementation in `version_2/models/fsa_high_res/_dynamics.py` uses an explicit *substepped Euler–Maruyama* scheme with boundary reflection. With outer step $\Delta t = 1/96$ days (15 minutes) and inner step count $n_{\text{sub}} = 4$, one outer step of the scheme reads

$$\begin{aligned} \tilde{y}_0 &= y_n, \\ \tilde{y}_{k+1} &= \tilde{y}_k + \frac{\Delta t}{n_{\text{sub}}}\, b(\tilde{y}_k, \Phi_n; \theta), \qquad k = 0, \ldots, n_{\text{sub}} - 1, \\ y^{\text{pre}}_{n+1} &= \tilde{y}_{n_{\text{sub}}} + \sigma(\tilde{y}_{n_{\text{sub}}}; \theta)\, \sqrt{\Delta t}\, \xi_n, \\ y_{n+1} &= \text{Reflect}(y^{\text{pre}}_{n+1}), \end{aligned} \tag{17}$$

where $b$ is the drift vector field, $\sigma$ the state-dependent diffusion vector, and $\xi_n \sim \mathcal{N}(0, I_3)$. The reflection operator enforces $B \in [0, 1]$ and $F, A \geq 0$ by mirroring any component that overshoots its boundary; in practice it almost never fires, because the diffusions $\sqrt{B(1-B)}, \sqrt{F}, \sqrt{A}$ vanish at their respective boundaries.

The substepping in the inner loop is purely a drift refinement: the Wiener increment is applied once per outer step. This separation of deterministic and stochastic time steps is convenient for the present stability analysis because the two contributions decouple.

## 5.2 What "numerical stability" means here

The strongest standard guarantee one can ask of an SDE solver is *mean-square stability*: that the second moment of the discrete-time iterate $E[\,|y_n|^2\,]$ remains bounded as $n \to \infty$ when the underlying continuous SDE is itself stable (in our case, exponentially ergodic via Theorem 8).

For a one-dimensional linear test SDE $\mathrm{d}X = aX\,\mathrm{d}t + bX\,\mathrm{d}W$, the explicit Euler–Maruyama iterate $X_{n+1} = X_n + aX_n\Delta t + bX_n\sqrt{\Delta t}\,\xi_n$ is mean-square stable if and only if

$$|1 + a\,\Delta t|^2 + b^2\,\Delta t \;\leq\; 1. \tag{18}$$

(See Higham, 2000.) For non-linear SDEs we apply (18) component-wise to the system linearised around its stable equilibria. This is the standard recipe and the one we follow.

### 5.3 Linearising the FSA-v2 SDE

Let $J = J(B^*, F^*, A^*)$ be the drift Jacobian at an equilibrium of the deterministic skeleton, and let $D = D(B^*, F^*, A^*)$ be the diagonal matrix of squared diffusion coefficients ($\sigma_B^2 B^*(1 - B^*)$, $\sigma_F^2 F^*$, $\sigma_A^2 A^*$). The mean-square stability constraint (18) applied component-wise gives, in the worst case,

$$\Delta t \;\leq\; \min_i \frac{2\,|\Re\lambda_i(J)|}{|\lambda_i(J)|^2 + \sigma_i^2(B^*, F^*, A^*)}, \tag{19}$$

where $\lambda_i(J)$ are the eigenvalues of $J$ and $\sigma_i^2$ are the corresponding diagonal entries of $D$. The slowest-decaying mode – the one that comes closest to the stability boundary – sets the binding constraint.

We worked out the eigenvalues of $J$ at the $A = 0$ branch in Section 4: they are $-1/\tau_B$, $-1/[\tau_F(1 + \lambda_A A_{\text{typ}})]$ and $\mu(B^*, F^*)$. With $\tau_B = 42$ days and $\tau_F^{\text{old}} = 7$ days, the slowest dissipative eigenvalue is $-1/\tau_B$, giving (very roughly, ignoring the diffusion contribution because $\sigma_B^2 B^*(1 - B^*) \leq \sigma_B^2/4$ is tiny):

$$\Delta t^B_{\text{stable}} \;\approx\; 2\tau_B \;=\; 84\,\text{d}. \tag{20}$$

For the $F$-mode the corresponding bound is $2\tau_F^{\text{old}} = 14$ days. On the oscillatory branch ($A \neq 0$) the binding eigenvalue is the cubic-damping rate $-2\eta(A^*)^2$, which at the limit-cycle amplitude $A^* \approx 1$ gives $\Delta t^A_{\text{stable}} \approx 1/\eta \approx 5\,\text{d}$. The full derivation, including Young's-inequality bounds that absorb the nonlinear corrections cleanly, is in Appendix C.

### 5.4 Stability margin at the current step size

The current outer step is $\Delta t = 1/96\,\text{d} \approx 0.010\,\text{d}$. Comparing to the bounds of Section 5.3,

$$\frac{\Delta t}{\Delta t^B_{\text{stable}}} \approx 1.2 \times 10^{-4}, \qquad \frac{\Delta t}{\Delta t^F_{\text{stable}}} \approx 7.4 \times 10^{-4}, \qquad \frac{\Delta t}{\Delta t^A_{\text{stable}}} \approx 2.1 \times 10^{-3}. \tag{21}$$

The current step size sits roughly *three orders of magnitude* inside the stability region. Numerical stability is therefore not the bottleneck and there is enormous head-room to enlarge $\Delta t$ before the solver itself becomes the failure mode.

### 5.5 What does bind, then?

Two practical considerations sit much closer to the current step size than the mean-square stability bound and should be the focus of any future enlargement of $\Delta t$.

**Resolution of the sub-daily Φ-burst.** The training-stimulus envelope of (7) is a Gamma($k = 2$) shape that peaks at $\tau_{\text{burst}} = 3\,\text{h}$ post-wake. Resolving such a peak with reasonable accuracy on a uniform time grid requires several time steps within the rise time, conservatively $\Delta t \leq \tau_{\text{burst}}/4 \approx 45\,\text{min}$. The current 15-minute step gives twelve grid points within the rise, i.e. a comfortable factor of three above the resolution floor.

**Strong convergence of the discretised paths.** The explicit Euler–Maruyama scheme converges with strong order 1/2 in the Itô sense. Pathwise error therefore scales as $\sqrt{\Delta t}$, and to halve the pathwise error one quarters the step. This is not a stability constraint, but it does cap the step size that retains useful pathwise fidelity for the inference work of Sections 7 and 8.

### 5.6 Recommendation

Combining the three constraints, the binding one for the present implementation is the resolution of the sub-daily $\Phi$-burst. Within that constraint, the step size could safely be enlarged from 15 to 30 minutes (still six grid points inside the rise of the Gamma envelope) at no cost to numerical stability and at modest cost to pathwise fidelity, halving the wall-clock cost of every SDE simulation in the code base.

A more aggressive enlargement to $\Delta t = 1\,\mathrm{h}$ would sit at the edge of the $\Phi$-burst-resolution constraint and would require a verification run – a drift-parity test against the 15-minute reference at the same truth point and the same Wiener realisation – before being adopted in production. We do not recommend going beyond 1 h without changing the solver: the substepped explicit scheme would remain stable, but the strong $\sqrt{\Delta t}$ pathwise-error growth would visibly degrade the filter posteriors.

*Remark* 10 (Improvements that would help). Three numerical-analysis upgrades would push the safe step size substantially higher and are flagged here for future work. (i) A Lamperti transform of the $F$- and $A$-CIR components would replace the $\sqrt{F}\,\mathrm{d}W$ and $\sqrt{A}\,\mathrm{d}W$ diffusions with constant-coefficient Brownian increments, removing the boundary reflection and the strong-error penalty near zero. (ii) A Milstein-type correction to the Euler–Maruyama step would lift the strong order from 1/2 to 1. (iii) An implicit treatment of the cubic damping term in the $A$-component would make the scheme A-stable in that direction, making the bound of Section 5.3 on the oscillatory branch unconditional. None of these are implemented; together they would plausibly support an outer step in the range $\Delta t \in [1, 4]\,\mathrm{h}$ with no degradation of inference quality, and would speed up the long-horizon Stage D and Stage E runs by an order of magnitude.

## 6 The bootstrap particle filter

### 6.1 Bayesian filtering recursion

Let $X_n \in \mathbb{R}^3$ denote the latent state $(B, F, A)$ at the $n$th fifteen-minute time step, and let $Y_n$ denote the corresponding multi-channel observation. For a fixed value of the static parameter $\theta_{\mathrm{dyn}}$, the FSA-v2 model defines a hidden Markov model with transition density

$$X_n \mid X_{n-1} \sim p(\,\cdot \mid X_{n-1},\, \Phi_n,\, \theta_{\mathrm{dyn}}) \tag{22}$$

implicitly through the Euler–Maruyama discretisation of (3)–(5), and observation density

$$Y_n \mid X_n \sim p(\,\cdot \mid X_n,\, C(t_n),\, \theta_{\mathrm{dyn}}) \tag{23}$$

given by the four-channel model (8)–(11). Bayesian filtering is the recursive computation of the conditional distribution $\pi_n(\mathrm{d}x_n) := p(\mathrm{d}x_n \mid Y_{1:n},\, \theta_{\mathrm{dyn}})$. It factorises into a one-step prediction and a Bayes update:

$$\pi_{n|n-1}(\mathrm{d}x_n) = \int p(\mathrm{d}x_n \mid x_{n-1},\, \Phi_n,\, \theta_{\mathrm{dyn}})\, \pi_{n-1}(\mathrm{d}x_{n-1}), \tag{24}$$

$$\pi_n(\mathrm{d}x_n) \propto p(Y_n \mid x_n,\, C(t_n),\, \theta_{\mathrm{dyn}})\; \pi_{n|n-1}(\mathrm{d}x_n). \tag{25}$$

Equations (24)–(25) have closed form only when both densities are linear-Gaussian (the Kalman case). For FSA-v2 they do not, and a Monte Carlo approximation is required.

### 6.2 Sequential importance resampling

The bootstrap particle filter (Gordon, Salmond and Smith, 1993; Doucet and Johansen, 2011) approximates $\pi_n$ by a weighted empirical measure

$$\pi_n^N(\mathrm{d}x_n) \;=\; \sum_{i=1}^{N} W_n^{(i)}\, \delta_{X_n^{(i)}}(\mathrm{d}x_n), \qquad \sum_{i=1}^{N} W_n^{(i)} = 1, \tag{26}$$

on a population of $N$ *state particles* $\{X_n^{(i)}\}$. The recursion has three substeps: (i) propagate each particle through the dynamics, (ii) reweight by the observation likelihood, (iii) resample to prevent weight degeneracy. This is sequential importance resampling (SIR), summarised in Algorithm 1. Resampling is performed *systematically*: with $u_0 \sim \mathcal{U}[0,1)$, the resampled indices are obtained by walking the cumulative weight $C(j) = \sum_{i \le j} W_n^{(i)}$ at the offsets $u_k = (k + u_0)/N$ for $k = 0, \dots, N-1$. Systematic resampling is $O(N)$, has lower Monte Carlo variance than multinomial resampling, and is deterministic given $u_0$.

**Algorithm 1** Bootstrap SIR for the FSA-v2 filter, given $\theta_{\text{dyn}}$

1: Sample $X_0^{(i)} \sim p(\mathrm{d}x_0 \mid \theta_{\text{dyn}})$ for $i = 1, \dots, N$; set $W_0^{(i)} = 1/N$.
2: **for** $n = 1, 2, \dots$ **do**
3: **Propagate.** For each $i$, draw $X_n^{(i)} \sim p(\cdot \mid X_{n-1}^{(i)}, \Phi_n, \theta_{\text{dyn}})$ by one Euler–Maruyama step of (3)–(5).
4: **Reweight.** Set $\tilde{W}_n^{(i)} = W_{n-1}^{(i)}\, p(Y_n \mid X_n^{(i)}, C(t_n), \theta_{\text{dyn}})$ and $W_n^{(i)} = \tilde{W}_n^{(i)} / \sum_j \tilde{W}_n^{(j)}$.
5: **ESS check.** Compute $N_{\text{eff}} = 1/\sum_i (W_n^{(i)})^2$. If $N_{\text{eff}} < N_{\text{thr}}$, resample $\{X_n^{(i)}\}$ systematically by $\{W_n^{(i)}\}$ and reset $W_n^{(i)} \leftarrow 1/N$.
6: **end for**

The implementation in `smc2fc/filtering/gk_dpf_v3_lite.py` adds two refinements that we mention but do not develop here. First, when parameter particles are coupled to the state particles in the SMC$^2$ outer layer of Section 7, a Liu–West shrinkage move is applied with an effective-sample-size dependent bandwidth, slowing parameter-cloud collapse. Second, when the effective sample size collapses despite resampling, an optional low-rank Sinkhorn optimal transport step blends a transport plan into the systematic resample. Both are diagnostic / numerical safeguards rather than changes to the mathematical statement of Algorithm 1.

### 6.3 The unbiased marginal likelihood

A key property of the bootstrap filter, which makes the SMC$^2$ algorithm of the next section possible, is that it produces an unbiased estimator of the marginal likelihood. With the unnormalised weights $\tilde{W}_n^{(i)}$ from line 4 of Algorithm 1, define

$$\widehat{L}_N(\theta_{\text{dyn}}) \;:=\; \prod_{n=1}^{T} \Big( \frac{1}{N} \sum_{i=1}^{N} \tilde{W}_n^{(i)} \Big). \tag{27}$$

**Proposition 11** (Unbiasedness; Del Moral, 2004)**.** *For any $N \ge 1$ and any $T \ge 1$, $\mathbb{E}[\,\widehat{L}_N(\theta_{\text{dyn}})\,] = p(Y_{1:T} \mid \theta_{\text{dyn}})$.*

A proof is in Del Moral (2004, Thm. 7.4.2). For the present notes the significance of Proposition 11 is operational: it allows us to plug $\widehat{L}_N(\theta_{\text{dyn}})$ in place of the (intractable) exact likelihood $p(Y_{1:T} \mid \theta_{\text{dyn}})$ inside any sampler over $\theta_{\text{dyn}}$ and obtain a procedure that targets exactly the right posterior. This is precisely what SMC$^2$ does.

## 7 SMC$^2$ over parameters

### 7.1 The tempered posterior over $\theta_{\text{dyn}}$

The bootstrap filter of Section 6 assumes the static parameter $\theta_{\text{dyn}}$ is known. In practice $\theta_{\text{dyn}}$ is itself the principal object of inference, with a prior $p(\theta_{\text{dyn}})$ and posterior

$$p(\theta_{\text{dyn}} \mid Y_{1:T}) \;\propto\; p(\theta_{\text{dyn}})\, p(Y_{1:T} \mid \theta_{\text{dyn}}). \tag{28}$$

A direct sampler is unavailable because $p(Y_{1:T} \mid \theta_{\text{dyn}})$ has no closed form. The SMC$^2$ algorithm of Chopin, Jacob and Papaspiliopoulos (2013) replaces this likelihood by the unbiased estimator $\widehat{L}_N(\theta_{\text{dyn}})$ from (27) and, for a fixed schedule $0 = \lambda_0 < \lambda_1 < \cdots < \lambda_K = 1$, defines a sequence of tempered targets

$$\pi_\lambda(\theta_{\text{dyn}}) \;\propto\; p(\theta_{\text{dyn}})^{1-\lambda}\, \widehat{L}_N(\theta_{\text{dyn}})^{\lambda}, \qquad \lambda \in [0,1]. \tag{29}$$

At $\lambda = 0$ the target is the prior; at $\lambda = 1$ it is, in expectation, the posterior. The interpolation in between bridges the two in a way that keeps the importance weights well-behaved.

### 7.2 Adaptive tempering and the move step

A population of $M$ *parameter particles* $\{\theta_{\text{dyn}}^{(m)}\}_{m=1,\dots,M}$ is propagated through the schedule $\{\lambda_k\}$. At each step the unnormalised tempering weight

$$\omega_k^{(m)} \;=\; \left(\frac{\widehat{L}_N(\theta_{\text{dyn}}^{(m)})}{p(\theta_{\text{dyn}}^{(m)})^{-1}}\right)^{\lambda_k - \lambda_{k-1}} \tag{30}$$

is computed, and an effective sample size $M_{\text{eff}} = (\sum_m \omega_k^{(m)})^2 / \sum_m (\omega_k^{(m)})^2$ is monitored. The interval $\lambda_k - \lambda_{k-1}$ is chosen *adaptively*, by bisection, to drive $M_{\text{eff}}/M$ to a prescribed target (typically 0.5); this avoids both unnecessarily short bridges and weight collapse. After resampling, each parameter particle is moved by a Hamiltonian Monte Carlo (HMC) kernel that targets $\pi_{\lambda_k}$, with the mass matrix re-estimated from the current particle cloud. Algorithm 2 summarises the procedure.

**Algorithm 2** Adaptive-tempered SMC$^2$ over $\theta_{\text{dyn}}$

Sample $\theta_{\text{dyn}}^{(m)} \sim p(\theta_{\text{dyn}})$ for $m = 1, \dots, M$; set $\lambda_0 = 0$.
For each $m$, run Algorithm 1 with $\theta_{\text{dyn}}^{(m)}$ to obtain $\widehat{L}_N(\theta_{\text{dyn}}^{(m)})$.
**while** $\lambda_k < 1$ **do**
    Solve for the smallest increment $\delta \in (0, 1-\lambda_k]$ such that the tempering ESS ratio with weights $\omega^{(m)} \propto \widehat{L}_N(\theta_{\text{dyn}}^{(m)})^{\delta}$ equals the target (e.g. 0.5). Set $\lambda_{k+1} = \lambda_k + \delta$.
    Resample $\{\theta_{\text{dyn}}^{(m)}\}$ systematically by $\{\omega^{(m)}\}$.
    **Move.** Run a small number of HMC steps targeting $\pi_{\lambda_{k+1}}$ on each $\theta_{\text{dyn}}^{(m)}$, re-running Algorithm 1 as needed to evaluate the unbiased likelihood at proposed values.
    $k \leftarrow k + 1$.
**end while**

### 7.3 Rolling-window SMC$^2$ and the Schrödinger–Föllmer bridge

When data arrive sequentially, running Algorithm 2 from scratch over the full record at every new arrival is wasteful. The implementation in this code base instead operates over a sliding sequence of *windows*: with a fifteen-minute time grid, each window is one day long (96 bins), the stride between consecutive windows is twelve hours (48 bins), and the macrocycle of fourteen days yields $(14 \cdot 96 - 96)/48 + 1 = 27$ windows.

For window $w \geq 2$ the parameter cloud is *warm-started* from the posterior of window $w - 1$. The transition is mediated by a Schrödinger–Föllmer bridge between two Gaussian fits, parameterised along the Bures–Wasserstein geodesic in the manifold of positive-definite covariance matrices. Concretely, let $(m_0, \Sigma_0)$ denote a Gaussian moment match to the previous window's posterior, and let $(m_1, \Sigma_1)$ denote an importance-weighted moment match obtained by a single particle-filter evaluation per particle on the new window's data. For a blend parameter

$\rho \in [0,1]$ the bridge measure has mean and covariance

$$m_\rho = (1-\rho)\, m_0 + \rho\, m_1, \tag{31}$$

$$\Sigma_\rho = \big((1-\rho)I + \rho\, T\big)\, \Sigma_0\, \big((1-\rho)I + \rho\, T\big)^\top, \tag{32}$$

where $T = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2}$ is the Bures–Wasserstein optimal-transport map. The default implementation uses $\rho = 0.7$, biasing the warm-start toward the new window's evidence. We do not develop the geometry here; the reader unfamiliar with it can treat the bridge as a black box that produces a sensible Gaussian initialisation for the next window's tempering schedule.

# 8 Control as inference

## 8.1 The cost functional

Sections 6 and 7 treat the dynamics parameters $\theta_{\text{dyn}}$ and the latent state $X$ as the things to be inferred from a fixed schedule $\Phi$. We now reverse the perspective: the dynamics parameters and the initial state are taken as given, and the schedule itself becomes the unknown.

For a fixed truth $\theta_{\text{dyn}}$, an initial state $x_0$, and a candidate schedule $\Phi : [0,T] \to [0, \Phi_{\max}]$, define the trajectory cost

$$J(\Phi) \;=\; \mathbb{E}_{X|\Phi}\left[ -\int_0^T A(t)\,\mathrm{d}t \;+\; \lambda_F \int_0^T \max\big(F(t) - F_{\max},\, 0\big)^2\,\mathrm{d}t \right], \tag{33}$$

where the expectation is over the law of $X = (B, F, A)$ under the SDE (3)–(5) driven by $\Phi$. The first term rewards mean autonomic amplitude; the second is a soft barrier penalising trajectories that exceed the fatigue ceiling $F_{\max} = 0.40$. The barrier weight is $\lambda_F = 1$ in the reference implementation; no $\Phi^2$ control penalty is included.

## 8.2 Schedule decoder

The schedule $\Phi$ is parameterised in finite dimensions by a vector $\theta_{\text{ctrl}} \in \mathbb{R}^8$ of Gaussian-RBF anchor coefficients. With a fixed basis $\{\varphi_a(t)\}_{a=1}^8$ (eight evenly spaced Gaussians on $[0,T]$ at the discretisation grid) and a logit bias $c_\Phi := \log\big(\Phi_{\text{def}}/(\Phi_{\max} - \Phi_{\text{def}})\big)$ chosen so that $\theta_{\text{ctrl}} = 0$ produces the canonical-Banister default $\Phi \equiv \Phi_{\text{def}} = 1.0$, the decoder is

$$\Phi(t;\, \theta_{\text{ctrl}}) \;=\; \Phi_{\max} \cdot \sigma\!\left( c_\Phi + \sum_{a=1}^{8} (\theta_{\text{ctrl}})_a\, \varphi_a(t) \right). \tag{34}$$

The sigmoid envelope keeps $\Phi(t;\, \theta_{\text{ctrl}}) \in [0, \Phi_{\max}]$ for every $\theta_{\text{ctrl}}$; the search is unconstrained in $\mathbb{R}^8$.

## 8.3 Tempered control posterior

Following the control-as-inference tradition (Kappen, 2005; Toussaint, 2009; Levine, 2018), we define a probability measure on $\theta_{\text{ctrl}}$ by treating the negative cost as an unnormalised log-likelihood. With inverse temperature $\beta > 0$ and a Gaussian prior $\theta_{\text{ctrl}} \sim \mathcal{N}(0,\, \sigma^2_{\text{prior}} I_8)$, the *control posterior* is

$$q_\beta(\theta_{\text{ctrl}}) \;\propto\; \exp\big(-\beta\, J(\theta_{\text{ctrl}})\big)\, \mathcal{N}(\theta_{\text{ctrl}};\, 0,\, \sigma^2_{\text{prior}} I_8), \tag{35}$$

where we now write $J(\theta_{\text{ctrl}}) := J(\Phi(\,\cdot\,; \theta_{\text{ctrl}}))$. Sampling from $q_\beta$ is performed by exactly the SMC$^2$ machinery of Section 7, but with the unbiased filter likelihood $\widehat{L}_N(\theta_{\text{dyn}})$ replaced by the cost-likelihood $\exp(-\beta\, J(\theta_{\text{ctrl}}))$. The inverse temperature $\beta$ is calibrated automatically from the cost spread of the prior cloud, and the tempered schedule $\lambda \in [0,1]$ smoothly interpolates from prior to control posterior.

### 8.4 Posterior mean as the controller output

**Proposition 12** (Posterior mean as a regularised cost minimiser)**.** *Let $q_\beta$ be the control posterior* (35) *and write $\bar{\theta} := \mathbb{E}_{q_\beta}[\theta_{\text{ctrl}}]$. Then*

$$\bar{\theta} \;=\; \arg\min_{\theta \in \mathbb{R}^8} \Big\{ \mathbb{E}_{q_\beta}\big[\, \|\theta - \theta_{\text{ctrl}}\|_2^2 \,\big] \Big\}, \tag{36}$$

*i.e. the posterior mean is the $L^2$ minimiser of squared distance under the tempered measure. In particular, when $J$ is approximately quadratic near its minimum and the prior variance is large enough that the prior contribution to $\nabla \log q_\beta$ is small, $\bar{\theta}$ sits in a neighbourhood of the unregularised minimiser $\arg\min_\theta J(\theta)$ of size $O(\beta^{-1/2})$.*

The implementation uses $\bar{\theta}$, computed as the equally-weighted mean of the SMC$^2$ parameter cloud at $\lambda = 1$, as the controller output. The corresponding schedule is then $\Phi(\,\cdot\,;\bar{\theta})$ via (34). We will see in Section 10 that this point output is sufficient in practice, despite throwing away the higher-order moments of $q_\beta$; Section 9.7 discusses why.

## 9 Receding-horizon model-predictive control

This section is the conceptual core of the document. It develops the finite-horizon stochastic optimal control problem, the receding-horizon principle, the various stochastic and output-feedback variants, and then specialises everything to the FSA-v2 closed-loop model-predictive controller in both mathematical and implementation form. A closing subsection sketches a linear–quadratic–Gaussian approximation that could be added as a cheap baseline.

### 9.1 Finite-horizon stochastic optimal control

Consider a discrete-time stochastic system on a state $X_k \in \mathcal{X}$ and a control $U_k \in \mathcal{U}$, with transition kernel $X_{k+1} \sim p(\,\cdot \mid X_k, U_k)$, a stage cost $\ell : \mathcal{X} \times \mathcal{U} \to \mathbb{R}$, and a terminal cost $V_f : \mathcal{X} \to \mathbb{R}$. The *open-loop* finite-horizon problem of length $H$ is

$$\min_{u_{0:H-1} \in \mathcal{U}^H} \; \mathbb{E}\left[ \sum_{k=0}^{H-1} \ell(X_k, u_k) + V_f(X_H) \,\middle|\, X_0 = x \right], \tag{37}$$

with the expectation taken over the joint law of $X_{1:H}$ given the chosen sequence $u_{0:H-1}$. Denote its optimal value by $V_H(x)$ and a minimising sequence by $u^*_{0:H-1}(x)$. The mapping $x \mapsto u^*_0(x)$ is the open-loop optimal first-action policy.

For FSA-v2 the state $X_k = (B_k, F_k, A_k)$ lives in $\mathcal{X} = [0,1] \times [0,\infty)^2$, and the "control sequence" is the eight-dimensional RBF-coefficient vector $\theta_{\text{ctrl}}$ that parameterises the entire schedule $\Phi(\,\cdot\,;\theta_{\text{ctrl}})$ over $[0,T]$ via (34). The discrete-time abstract notation (37) therefore maps onto the continuous schedule of Section 8 by a one-to-one identification between admissible $u_{0:H-1}$ and admissible $\theta_{\text{ctrl}}$. The expectation in (37) is the same one already written out in (33).

### 9.2 The receding-horizon principle

In *receding-horizon* or *model-predictive* control, the open-loop problem (37) is solved repeatedly on a sliding window. At each replan time $t_n$, given a current state estimate $\widehat{x}_n$, one

1. solves (37) for $u^*_{0:H-1}(\widehat{x}_n)$;
2. applies only the first action $u^*_0(\widehat{x}_n)$ over a stride of length $\Delta < H$;
3. observes new data, updates the state estimate to $\widehat{x}_{n+1}$, and returns to step 1.

The resulting closed-loop policy is a function of $\widehat{x}_n$ alone, so the controller is implementable. Stability of the closed loop is delicate. Mayne and Rawlings (2000) show that under suitable assumptions on the horizon length and the terminal cost / terminal set, the closed-loop state remains in a positively invariant set and converges to a neighbourhood of a desired equilibrium. For FSA-v2 we rely on the *long-horizon* variant of that theory: the planning horizon $H$ is chosen long enough relative to the slowest time constant $\tau_B = 42$ days that the truncation error from neglecting trajectory beyond $H$ is small.

### 9.3 Stochastic and output-feedback variants

Three flavours of stochastic MPC are commonly distinguished.

(i) *Certainty-equivalent* MPC. Replace the expectation in (37) by evaluation at a point estimate $\hat{X} = \mathbb{E}[X \mid \widehat{x}_n]$, and solve the resulting deterministic problem. Cheap, but discards uncertainty information.

(ii) *Expected-cost* (stochastic) MPC. Keep the expectation, and approximate it by Monte Carlo (sampling trajectories of the stochastic dynamics) or by scenario trees.

(iii) *Output-feedback* MPC. When $X_k$ is not observed directly, the controller acts on a state estimate $\widehat{x}_n$ produced by an external estimator (Kalman, particle filter, SMC$^2$, etc.). For nonlinear non-Gaussian systems no *separation principle* holds in general — the joint design of the estimator and the controller is part of the modelling, not a free composition.

The FSA-v2 pipeline is in class (ii) and (iii) simultaneously. The state is unobserved (only the four-channel proxies of (8)–(11) are available), so a state estimator is required: this is the rolling-window SMC$^2$ filter of Section 7. The expectation in the cost (33) is kept, evaluated by Monte Carlo on the SDE under common random numbers, and minimised in $\theta_{\text{ctrl}}$-space by the control SMC$^2$ of Section 8. The full closed-loop policy is therefore the composition

$$(\text{observations}) \xrightarrow{\text{rolling SMC}^2\text{ filter}} \widehat{x}_n \xrightarrow{\text{control SMC}^2\text{, posterior mean}} \bar{\theta} \xrightarrow{(34)} \Phi(\,\cdot\,;\bar{\theta}) \xrightarrow{\text{plant}} (\text{new observations}).$$

There is no separation principle here. The fact that this composition works at all is an empirical claim, validated in Section 10.

### 9.4 The FSA-v2 MPC pipeline — mathematical statement

We now write down the closed-loop algorithm. Let $\Delta_w = 96$ bins be the filter window length (one day), $\Delta_s = 48$ bins the stride (twelve hours), and $H_T = 14 \cdot 96 = 1344$ bins the macrocycle (fourteen days), so that the number of windows is $N_w = (H_T - \Delta_w)/\Delta_s + 1 = 27$.

**Algorithm 3** FSA-v2 closed-loop MPC

1: Initialise the plant at $(B_0, F_0, A_0)$; initialise an empty observation buffer $Y_{1:0} = \emptyset$.
2: **for** $n = 1, \ldots, N_w$ **do**
3: **Filter update.** If $n = 1$, run cold-start SMC$^2$ (Algorithm 2) on $Y_{(n-1)\Delta_s+1:(n-1)\Delta_s+\Delta_w}$ to produce a posterior over $\theta_{\text{dyn}}$ for the first window; otherwise warm-start from the previous window's posterior via the Schrödinger–Föllmer bridge of Section 7.
4: **State extraction.** For a small set of $\theta_{\text{dyn}}$ particles drawn from the posterior, run the bootstrap filter (Algorithm 1) forward to the stride endpoint and compute the importance-weighted mean of the resulting $(B, F, A)$; call this $\widehat{x}_n$.
5: **Control plan.** Run the control SMC$^2$ of Section 8, conditioned on $\widehat{x}_n$ and the posterior-mean dynamics parameters $\bar{\theta}_{\text{dyn}}$, to obtain the posterior-mean schedule coefficient $\bar{\theta}_{\text{ctrl}}$.
6: **Apply.** Decode $\Phi(\,\cdot\,;\bar{\theta}_{\text{ctrl}})$ via (34) and apply only its first stride (length $\Delta_s$) to the plant; collect the resulting observations and append them to the buffer.
7: **end for**

The horizon over which the controller plans (line 5) is the full macrocycle remainder $H_T - (n-1)\Delta_s$, but only the first $\Delta_s$ bins of the plan are committed (line 6). This is the receding-horizon truncation of Section 9.2 applied to the FSA-v2 problem. Replanning occurs every twelve hours, at the wake-window boundary so that the morning-loaded burst envelope of (7) is delivered in full each day.

### 9.5 Code-level structure of the MPC implementation

This subsection departs from the usual mathematical idiom and describes how the abstract loop of Section 9.4 is realised in the source code of the project. The reader who only wants the mathematics may skip to Section 9.6; the reader who intends to run, modify, or extend the controller will find this useful. We name three artefacts.

**(a) The plant interface, `StepwisePlant`.** The "plant" in stochastic optimal control is whatever object the controller acts upon to produce new observations. In the code base it is the class `StepwisePlant` in `version_2/models/fsa_high_res/_plant.py`. A `StepwisePlant` carries the hidden true state $(B, F, A)$, the true dynamics parameters, and a global bin counter; its single mutating method is

```
advance(stride_bins, Phi_daily) → (trajectory, obs_HR, obs_sleep, obs_stress,
obs_steps).
```

A call to `advance` expands the supplied daily-$\Phi$ array into the morning-loaded sub-daily envelope of (7) (preserving $\int_{\text{day}} \Phi \, \mathrm{d}t = 24\,\Phi_d$ exactly), runs Euler–Maruyama for `stride_bins` fifteen-minute steps, samples the four-channel observations of (8)–(11), and updates the internal state. Pedagogically, the `StepwisePlant` is the digital embodiment of the "real world" against which the controller is evaluated; mathematically, a single call samples one transition of the joint state-observation process under the chosen control over a stride of length $\Delta_s$.

**(b) The MPC driver, `bench_smc_full_mpc_fsa.py`.** The receding-horizon orchestrator lives in `version_2/tools/bench_smc_full_mpc_fsa.py`. Its execution sequence implements Algorithm 3 step by step. For each window $n$ it

1. runs the rolling-window filter by calling `run_smc_window` (cold-start, $n = 1$) or `run_smc_window_bridge` with a Schrödinger–Föllmer base measure (warm-start, $n \geq 2$);
2. extracts the posterior-mean dynamics parameters and the smoothed latent state $\widehat{x}_n$ at the end of the window, the latter computed by running Algorithm 1 forward on a small set of $\theta_{\text{dyn}}$ particles drawn from the posterior and taking the weighted mean of the resulting $(B, F, A)$ samples;
3. assembles a `ControlSpec` (cost functional, horizon length, current state estimate, RBF basis);
4. calls the control SMC$^2$ loop in `smc2fc/control/tempered_smc_loop.py`, which returns a posterior-mean schedule coefficient $\bar{\theta}_{\text{ctrl}}$ and a corresponding $\Phi$ schedule;
5. calls `plant.advance(stride_bins, Phi_daily)` to apply only the first stride of that schedule;
6. appends the resulting observations to the data buffer and shifts the window.

**(c) The posterior-mean handoff between windows.** The smoothed state $\widehat{x}_n$ extracted in step 2 above is what initialises the next window's particle filter. Consecutive windows are therefore connected *at the state level* by a single point estimate (the importance-weighted smoothed mean) and *at the parameter level* by the SF Bures–Wasserstein bridge of Section 7. This is the practical realisation of the warm-start that gives the rolling SMC$^2$ its computational tractability: at fourteen days of fifteen-minute data the cold-start cost would be 27× larger.

Table 4 collects the mapping between the steps of Algorithm 3 and the source files that realise them.

| Step in Algorithm 3 | Concept | Source |
|---|---|---|
| 3 | rolling SMC$^2$ filter | `smc2fc/core/tempered_smc.py` |
| 3 | SF Bures–Wasserstein bridge | `smc2fc/core/sf_bridge.py` |
| 3 | inner bootstrap filter | `smc2fc/filtering/gk_dpf_v3_lite.py` |
| 4 | smoothed-state extraction | `version_2/tools/bench_smc_full_mpc_fsa.py` |
| 5 | control SMC$^2$ outer loop | `smc2fc/control/tempered_smc_loop.py` |
| 5 | cost functional, RBF decoder | `version_1/models/fsa_high_res/control.py` |
| 6 | plant advance | `version_2/models/fsa_high_res/_plant.py` |
| 2,6 | sub-daily $\Phi$ envelope | `version_2/models/fsa_high_res/_phi_burst.py` |

Table 4: Mapping from Algorithm 3 to the source code.

### 9.6 Recursive feasibility and the F-barrier

The fatigue ceiling $F_{\max} = 0.40$ is enforced through the soft-barrier penalty $\lambda_F \cdot \max(F - F_{\max}, 0)^2$ in the cost (33), not through a hard constraint. The choice is deliberate. Under stochastic dynamics with state-dependent diffusion and unbounded driving noise, an exact hard constraint $F(t) \leq F_{\max}$ cannot be guaranteed by any controller without making the admissible-control set empty for some achievable initial conditions; this is the standard *recursive feasibility* difficulty in stochastic MPC. A soft penalty preserves recursive feasibility – there is always a feasible schedule – and produces the constraint approximately, with the violation rate decreasing as $\lambda_F$ is increased. Empirically (see Section 10) the closed-loop F-violation rate is 1.28% on the 84-day open-loop benchmark and 0.00% on the 14-day closed-loop benchmark, well within the 5% acceptance gate.

### 9.7 Robustness and the flat-cost-surface argument

The controller of Section 9.4 commits the posterior mean $\bar{\theta}_{\text{ctrl}}$ rather than a sample from the control posterior or the full distribution. One might worry that this discards useful information, especially when the filter posterior over $\theta_{\text{dyn}}$ is uncertain or biased. Empirically the policy is nevertheless robust to filter parameter drift, and the explanation is geometric: near the optimum, the cost $J(\theta_{\text{ctrl}})$ is locally flat in the directions explored by the SMC$^2$ cloud. A first-order Taylor expansion of $J$ at the posterior mean gives, for a perturbed estimate $\bar{\theta} + \delta$,

$$J(\bar{\theta} + \delta) \;\approx\; J(\bar{\theta}) + \tfrac{1}{2}\delta^\top H \delta, \qquad H = \nabla^2 J(\bar{\theta}), \tag{38}$$

so the additional cost from using $\bar{\theta}$ in place of an unknown "oracle" optimum scales with the Hessian. When $H$ is small in the directions of greatest filter uncertainty, the closed-loop performance is correspondingly insensitive to filter accuracy. Section 10 reports closed-loop performance on a record where only 3 out of 26 estimable parameters pass a strict coverage gate; the controller nevertheless achieves a +17% amplitude gain. The flat-cost-surface argument is the mechanism.

### 9.8 An exact-approximate alternative: LQG and the differential Riccati equation

We close Section 9 by sketching a second controller that could be added to the project as a cheap baseline or as a warm-start for the SMC$^2$ control of Section 8. The sketch is deliberately complete enough to make the assumptions and limitations explicit; it is not yet implemented in the code base, and is flagged as required next work in Section 11.3.

**Linearisation around an operating point.** Three assumptions reduce the FSA-v2 problem to one with a closed-form optimal feedback law.

**Assumption 13.** There is a nominal trajectory $(x^*(t), \Phi^*(t))$ on $[0, T]$, taken in the simplest case to be the deterministic-skeleton equilibrium $(B^*, F^*, A^*)(\Phi_{\text{const}})$ under a constant control. With $\tilde{x} = x - x^*$ and $\tilde{\Phi} = \Phi - \Phi^*$, define the linearised drift coefficients

$$A_{\text{lin}}(t) = \left.\frac{\partial f}{\partial x}\right|_*, \qquad B_{\text{lin}}(t) = \left.\frac{\partial f}{\partial \Phi}\right|_*, \tag{39}$$

where $f$ is the drift of (3)–(5).

The cubic Stuart–Landau term $-\eta A^3$ is linearised away under Assumption 13, so the bistable / limit-cycle behaviour of the autonomic amplitude is lost. The linearisation is valid only in a neighbourhood of the operating point.

**Assumption 14.** The state-dependent Jacobi and CIR diffusions in (3)–(5) are replaced by constant-coefficient Gaussian noise $\bar{\sigma}\,\mathrm{d}W$ evaluated at the operating point.

The $[0, 1]$ and $[0, \infty)$ state constraints are no longer enforced implicitly by the noise vanishing at the boundary, so trajectories may leave the physiological domain. Acceptable when fluctuations around the operating point are small relative to the boundary distances.

**Assumption 15.** The cost (33) is replaced by the quadratic surrogate

$$J_{\text{LQ}}(\tilde{\Phi}) \;=\; \mathbb{E}\left[\int_0^T (\tilde{x}^\top Q \tilde{x} + \tilde{\Phi}^\top R \tilde{\Phi})\,\mathrm{d}t + \tilde{x}(T)^\top Q_T \tilde{x}(T)\right], \tag{40}$$

with $Q \succeq 0$, $R \succ 0$, $Q_T \succeq 0$.

To approximate the $-\int A\,\mathrm{d}t$ reward, one chooses $x_{\text{ref}} \neq x^*$ shifted toward higher $A$ and penalises $\tilde{x} = x - x_{\text{ref}}$ with $Q$ giving most weight to the $A$ component; this preserves $Q \succeq 0$. The soft fatigue barrier $\lambda_F \cdot \max(F - F_{\max}, 0)^2$ is replaced by a quadratic in $F - F^*$, which penalises both directions symmetrically – too restrictive on the rest side and too lax on the over-fatigue side near the barrier.

**The differential Riccati equation.** Under Assumptions 13–15, the optimal feedback law for the linearised system is the time-varying linear policy

$$\tilde{\Phi}^*(t) \;=\; -K(t)\,\tilde{x}(t), \qquad K(t) \;=\; R^{-1}\,B_{\text{lin}}(t)^\top\,P(t), \tag{41}$$

where $P : [0, T] \to \mathbb{R}^{3\times 3}$, $P(t) \succeq 0$, solves the *differential Riccati equation*

$$-\dot{P}(t) \;=\; A_{\text{lin}}(t)^\top P(t) + P(t)\,A_{\text{lin}}(t) - P(t)\,B_{\text{lin}}(t)\,R^{-1}\,B_{\text{lin}}(t)^\top P(t) + Q, \qquad P(T) = Q_T. \tag{42}$$

For a time-invariant linearisation (constant $A_{\text{lin}}$, $B_{\text{lin}}$) and an infinite horizon, $P$ solves the *algebraic* Riccati equation $0 = A_{\text{lin}}^\top P + P\,A_{\text{lin}} - P\,B_{\text{lin}}\,R^{-1}\,B_{\text{lin}}^\top P + Q$. Under Assumptions 13–14 the state-estimation problem also linearises and is solved by the Kalman–Bucy filter; the LQG controller is then the certainty-equivalent composition of Kalman–Bucy filter and LQR feedback, and the (linear-Gaussian) separation principle holds.

**Where this would fit in the project.** A complete LQG companion to the SMC$^2$ controller would consist of a new module `smc2fc/control/lqg/` containing

1. automatic linearisation of the drift in `version_2/models/fsa_high_res/_dynamics.py` around a user-supplied operating point, via JAX automatic differentiation;
2. a numerical solver for (42) by backward integration on the same fifteen-minute time grid used elsewhere in the project;
3. a `LQGController` class emitting the feedback gain $K(t)$ of (41) and a corresponding $\Phi$ schedule on the project's time grid.

The natural uses are (i) as a *cheap baseline* against which the SMC$^2$ controller of Section 8 is benchmarked — does the non-linear posterior-mean controller actually outperform the linear LQG approximation, and by how much? — and (ii) as a *warm-start* for the control posterior, by least-squares projection of the LQG schedule onto the RBF basis to give an initial $\theta_{\text{ctrl}}$ around which the SMC$^2$ then explores. None of this currently exists in the repository; we return to it in Section 11.3.

# 10 Results

The two headline empirical results of the project are an open-loop benchmark on the canonical 84-day horizon (one chronic time constant of 42 days, doubled) and the closed-loop fourteen-day rolling controller. Both compare the SMC$^2$ schedule to the canonical constant baseline $\Phi \equiv 1.0$. Headline numbers are summarised in Table 5.

| Benchmark | Horizon | $\bar{A}$ (SMC$^2$) | $\bar{A}$ (baseline) | gain | F-violation |
|---|---|---|---|---|---|
| Open-loop, Stage D | 84 d | 0.645 | 0.503 | +28% | 1.28% |
| Closed-loop MPC, Stage E5 | 14 d | 0.0964 | 0.0823 | +17% | 0.00% |

Table 5: Headline empirical results. $\bar{A}$ denotes the mean autonomic amplitude $\frac{1}{T}\int_0^T A(t)\,\mathrm{d}t$ averaged over Monte Carlo trajectories. Sources: `version_1/outputs/fsa_high_res/RESULT.md` (Stage D), `version_2/outputs/fsa_high_res/RESULT.md` (Stage E5).

## 10.1 Stage D: open-loop schedule on the 84-day horizon

In the open-loop benchmark, the dynamics parameters $\theta_{\text{dyn}}$ are treated as known at their truth values, the initial state is fixed at $(B_0, F_0, A_0) = (0.05, 0.30, 0.10)$ (a de-trained athlete with high residual fatigue), and the SMC$^2$ control of Section 8 is run *once* over the full 84-day horizon. The output is a single posterior-mean schedule $\Phi(\,\cdot\,;\bar{\theta})$.

The optimised schedule has mean $\Phi \approx 1.96$ over the horizon and a front-loaded periodisation profile (heavy training for the first $\sim 60$ days, taper thereafter). Mean amplitude $\bar{A} = 0.645$ exceeds the constant baseline $\bar{A} = 0.503$ by 28%, exceeds the sedentary reference $\bar{A} = 0.209$ by a factor of 3.08, and the soft fatigue barrier is violated only 1.28% of the time — well within the 5% acceptance gate of Section 9.6. All four acceptance gates (mean-A matches baseline within 3%, mean-A $\geq 1.40\times$ sedentary, mean-$\Phi \in [0.5, 2.5]$, F-violation $\leq 5\%$) pass.

## 10.2 Stage E5: closed-loop MPC on the 14-day macrocycle

In the closed-loop benchmark the dynamics parameters are no longer known to the controller. Algorithm 3 is run on a 14-day record of four-channel observations, with a one-day filter window, twelve-hour stride, and 27 replans. The plant is the `StepwisePlant` of Section 9.5.

Mean amplitude under the closed-loop policy is $\bar{A} = 0.0964$, against $\bar{A} = 0.0823$ for the constant baseline $\Phi \equiv 1.0$ on the same 14-day horizon. (The shorter horizon explains the smaller absolute numbers compared to Stage D: the integral $\int_0^T A(t)\,\mathrm{d}t$ has not yet converged to the asymptotic mean.) The +17% gain is achieved with a remarkable schedule shape: the controller commits a *rest-leaning* $\Phi$ in the range 0.25–0.31 across all 27 strides, against the canonical baseline of 1.0. Physiologically, this is the controller exploiting the fact that the simulated athlete starts with high residual fatigue ($F_0 = 0.30$): the optimal short-horizon strategy is to clear fatigue rather than to add training stimulus, which unlocks the Stuart–Landau growth-rate $\mu(B, F)$ and lets $A$ rise from 0.10 to $\sim 0.20$ over the macrocycle.

The fatigue ceiling is never violated (0.00%). Wall-clock cost on a single GPU is 41 minutes for the full closed loop. The closed-loop policy is robust despite imperfect filter coverage — on this record only 3 of 26 estimable parameters meet a strict $\geq 5/6$ coverage gate at the window level — consistent with the flat-cost-surface argument of Section 9.7.

A trajectory plot of $A(t)$ under both policies, and a plot of the SMC$^2$ posterior-mean schedule, are included as Figure placeholders here; they can be regenerated from the artefacts in `version_2/outputs/fsa_high_res/` by re-running `tools/bench_smc_full_mpc_fsa.py`.

*Figure placeholder: $A(t)$ trajectory under closed-loop MPC (blue) and constant baseline $\Phi \equiv 1$ (grey), with the discovered rest-leaning $\Phi$ schedule overlaid.*

# 11 Discussion and open questions

The empirical results of Section 10 show that the SMC$^2$ filter and controller, composed in receding-horizon form, produce a closed-loop policy that improves mean autonomic amplitude by +17% over a sensible constant baseline on the FSA-v2 model, with no fatigue-ceiling violations.

Earlier drafts of these notes flagged three analyses as outstanding: Fisher information rank identifiability, Lyapunov stability, and a linear–quadratic–Gaussian (LQG) approximation. The first two have since been carried out and now form Sections 3 and 4. The third remains as the principal open piece of analytical work and is described in Section 11.3 below. Three smaller open questions are then collected in Section 11.4.

## 11.1 Resolved: Fisher information rank identifiability

The pen-and-paper FIM analysis of Section 3 identified three rank-deficient pairs in the original v2 parametrization, $(\kappa_B, \varepsilon_A)$, $(\mu_F, \mu_{FF})$, and $(\tau_F, \lambda_A)$. Each pair was shown to enter the one-day-window likelihood only through a single linear combination, which is the combinatorial signature of a rank-one rather than rank-two contribution to the FIM. The Stage G1 reparametrization rotates each pair into a strongly-identified combination and a residual that vanishes at the operating point of Definition 1; empirically this restores full FIM rank and contracts the posterior standard deviations on the residuals by roughly a factor of six (see Table 3).

A useful follow-up, identified in Remark 4, is to add a numerical FIM module under `smc2fc/diagnostics/fim/` that recomputes the rank and condition number directly from the JAX dynamics, so that any future modification of the model triggers an automatic re-check.

## 11.2 Resolved: Lyapunov stability of the SDE

Section 4 carried out the stability analysis in two parts. The deterministic skeleton, under a constant control $\Phi \equiv \Phi_c$, has an explicit $A = 0$ equilibrium whose local stability is determined entirely by the sign of the Stuart–Landau growth rate $\mu(B_0^*, F_0^*)$ (Proposition 5); the bifurcation locus of Section 4.1 crosses zero at $\Phi_c \approx 7.5$, above which overtraining-collapse renders the $A = 0$ branch stable. For the full SDE we exhibited a quadratic Lyapunov function $V =$

$\frac{1}{2}(aB^2+bF^2+cA^2)$ that satisfies a Foster–Lyapunov drift inequality (Theorem 7), and concluded exponential ergodicity at rate $\rho < \alpha = \min(2/\tau_B, 1/\tau_F^{\text{old}})$ (Theorem 8). This delivers strong stability of the controlled process on the operating domain $\mathcal{X} = [0,1] \times \mathbb{R}_+^2$ for any bounded control schedule.

Remark 9 noted that the constants in Theorem 7 are loose: they use the simplest quadratic Lyapunov function and global bounds on the residual factors. A sharper rate could be obtained from a piecewise Lyapunov function that switches between the $\mu < 0$ and $\mu > 0$ regimes identified in Proposition 5, or from a non-quadratic Lyapunov function tuned to the geometry of the Stuart–Landau cubic.

### 11.3 Outstanding: LQG / Riccati exact-approximate alternative

Section 9.8 sketched a linear–quadratic–Gaussian controller for FSA-v2 under three explicit assumptions:

(A1) linearisation of the drift around an operating point;

(A2) replacement of the state-dependent Jacobi/CIR diffusions by constant-coefficient Gaussian noise;

(A3) replacement of the asymmetric $-\int A\,\mathrm{d}t$ objective and the soft fatigue barrier by a quadratic surrogate cost.

Under (A1)–(A3), the optimal feedback law is the time-varying linear policy (41) with gain matrix $K(t) = R^{-1}B_{\text{lin}}(t)^\top P(t)$, where $P(t)$ solves the differential Riccati equation (42). None of this is yet implemented in the repository.

We identify three concrete pieces of work that would deliver an LQG companion to the SMC$^2$ controller:

1. Implement automatic linearisation of the drift around a user-supplied operating point, using the existing JAX dynamics in `version_2/models/fsa_high_res/_dynamics.py`; expose the time-varying $A_{\text{lin}}(t), B_{\text{lin}}(t)$ of (39) on the project's fifteen-minute grid.

2. Implement a backward-integration solver for the differential Riccati equation (42).

3. Build an `LQGController` that emits the feedback gain $K(t)$ of (41) and a corresponding $\Phi$ schedule.

Two scientific questions then become accessible. *Does the LQG schedule, evaluated as an open-loop control on the FSA-v2 model, achieve mean amplitude comparable to the SMC$^2$ schedule on the 84-day benchmark of Section 10?* If yes, the non-linear SMC$^2$ machinery may be unnecessary in steady-state regimes; if no, the gap quantifies the value of capturing the cubic Stuart–Landau non-linearity. *Does using the LQG schedule as a warm-start for the SMC$^2$ control of Section 8 reduce wall-clock time to solution?* A reasonable hypothesis is that LQG provides an excellent initial $\theta_{\text{ctrl}}$ in the linear regime and only marginal benefit where the non-linearity matters; experimentally testing this is the natural next step.

### 11.4 Other open questions

A few smaller points deserve mention but are not, in the authors' view, on the same level of importance as the LQG analysis above.

- *Horizon length.* The Stage E5 result uses a fourteen-day macrocycle. The Stage D open-loop result uses eighty-four days. It is not yet known how the closed-loop gain scales with horizon length; a horizon sweep is straightforward to set up.

- *Posterior shape vs. posterior mean.* The controller of Section 9.4 uses only the posterior mean $\bar{\theta}_{\text{ctrl}}$. The flat-cost-surface argument of Section 9.7 explains why this is sometimes sufficient, but in regimes where the cost surface is curved the full posterior carries useful information. A risk-sensitive variant that minimises a quantile of the cost rather than the mean would be the natural extension.
- *Parameter drift across windows.* The Bures–Wasserstein bridge of Section 7 produces a sensible Gaussian-fit warm-start, but does not protect against *slow* parameter drift in the truth (e.g. the athlete's time constants change with training age). A more principled approach would augment the static parameters with a slow random-walk component and let the filter track the drift explicitly.

With the FIM and Lyapunov analyses now in place, the model has both empirical validation through the Stage D and Stage E5 results of Section 10 and analytical guarantees on identifiability and stability. The remaining theoretical work concentrates on the LQG / Riccati alternative.

# A Fisher information rank deficiencies of the pre-G1 model: derivations

This appendix supplies the derivations underlying the identifiability analysis of Section 3. We derive analytically the three rank-deficient parameter pairs of the pre-G1 FSA-v2 model (Sections A.1–A.3), describe the resulting null space of the FIM (Section A.4), and prove that the G1 reparametrization is mathematically equivalent to the pre-G1 model under a fixed relabelling of parameters (Section A.5).

Throughout, we use the operating point of Definition 1, write $\delta A = A - A_{\text{typ}}$ and $\delta F = F - F_{\text{typ}}$ for state fluctuations, and recall that at one-day windows the slow Banister state moves by at most $O(\mathrm{d}t/\tau)$ where $\tau \in \{\tau_B, \tau_F\}$, so $\delta A$ and $\delta F$ are small in this regime.

## A.1 Pair 1: $(\kappa_B, \varepsilon_A)$

The pre-G1 $B$-drift is

$$\mathrm{d}B = \big[\kappa_B(1+\varepsilon_A A)\Phi - B/\tau_B\big]\mathrm{d}t + \cdots$$

Linearising the factor $1+\varepsilon_A A$ around $A_{\text{typ}}$ and writing $\delta A = A - A_{\text{typ}}$,

$$\kappa_B(1+\varepsilon_A A)\,\Phi \;=\; \underbrace{\kappa_B(1+\varepsilon_A A_{\text{typ}})}_{=:\,\kappa_B^{\text{eff}}}\,\Phi \;+\; \kappa_B\,\varepsilon_A\,\delta A\,\Phi. \tag{43}$$

The first term carries the dependence of the daily drift on the parameters via the single product $\kappa_B^{\text{eff}}$; the second term is $O(\delta A)$, vanishing at the operating point.

Now consider how the data depend on small perturbations $(\delta\kappa_B, \delta\varepsilon_A)$. Differentiating (43) at the truth point gives

$$\frac{\partial}{\partial\kappa_B}\big[\kappa_B(1+\varepsilon_A A)\Phi\big] = (1+\varepsilon_A A)\,\Phi = (1+\varepsilon_A A_{\text{typ}})\,\Phi + \varepsilon_A\delta A\,\Phi,$$

$$\frac{\partial}{\partial\varepsilon_A}\big[\kappa_B(1+\varepsilon_A A)\Phi\big] = \kappa_B A\,\Phi = \kappa_B A_{\text{typ}}\,\Phi + \kappa_B\delta A\,\Phi.$$

At the operating point (where $\delta A = 0$) these two gradients are linearly dependent: their ratio is $(1+\varepsilon_A A_{\text{typ}})/(\kappa_B A_{\text{typ}})$, independent of $\Phi$. The score function with respect to $\kappa_B$ and the score function with respect to $\varepsilon_A$ are therefore proportional, with proportionality constant fixed by the truth values. Their outer product in (12) is rank one, not rank two.

The combination that the data *can* resolve is precisely the product $\kappa_B^{\text{eff}} = \kappa_B(1+\varepsilon_A A_{\text{typ}})$. The orthogonal direction $(\delta\kappa_B, \delta\varepsilon_A) \perp \nabla\kappa_B^{\text{eff}}$ moves the parameter vector but leaves the daily drift unchanged at the operating point.

### A.2 Pair 2: $(\mu_F, \mu_{FF})$

The pre-G1 Stuart–Landau growth rate is $\mu(B,F) = \mu_0 + \mu_B B - \mu_F F - \mu_{FF} F^2$. Completing the square in the quadratic in $F$,

$$\begin{aligned} -\mu_F F - \mu_{FF} F^2 &= -\mu_F F - \mu_{FF}\big[(F - F_{\text{typ}})^2 + 2F_{\text{typ}} F - F_{\text{typ}}^2\big] \\ &= -\underbrace{(\mu_F + 2F_{\text{typ}}\mu_{FF})}_{=:\,\mu_F^{\text{eff}}} F \;-\; \mu_{FF}(F - F_{\text{typ}})^2 + \mu_{FF} F_{\text{typ}}^2. \end{aligned} \tag{44}$$

The constant $\mu_{FF} F_{\text{typ}}^2$ shifts the intercept $\mu_0$ but is uninformative about $\mu_F$ or $\mu_{FF}$ separately; the linear-in-$F$ slope $\mu_F^{\text{eff}}$ does carry their combined effect; and the quadratic $-\mu_{FF}(F - F_{\text{typ}})^2$ is $O(\delta F^2)$, so it contributes at second order in the small fluctuations of $F$ at one-day windows.

Differentiating $\mu(B,F)$ with respect to $\mu_F$ and $\mu_{FF}$ at the truth point and setting $\delta F = 0$,

$$\partial_{\mu_F}\, \mu(B,F)\big|_{F_{\text{typ}}} = -F_{\text{typ}}, \qquad \partial_{\mu_{FF}}\, \mu(B,F)\big|_{F_{\text{typ}}} = -F_{\text{typ}}^2.$$

Both gradients point in the same one-dimensional subspace (they differ only by the scalar factor $F_{\text{typ}}$), so the FIM contribution from $(\mu_F, \mu_{FF})$ at the operating point is rank one. The combination resolved is $\mu_F^{\text{eff}} = \mu_F + 2F_{\text{typ}}\mu_{FF}$ (the slope at $F_{\text{typ}}$); the orthogonal direction is silent.

### A.3 Pair 3: $(\tau_F, \lambda_A)$

The pre-G1 $F$-clearance rate is $(1 + \lambda_A A)/\tau_F$. Linearising in $\delta A$,

$$\frac{1 + \lambda_A A}{\tau_F} \;=\; \frac{1 + \lambda_A A_{\text{typ}}}{\tau_F} + \frac{\lambda_A}{\tau_F}\,\delta A \;=:\; \frac{1}{\tau_F^{\text{eff}}} + O(\delta A), \qquad \tau_F^{\text{eff}} := \frac{\tau_F}{1 + \lambda_A A_{\text{typ}}}. \tag{45}$$

The leading term is the inverse effective acute timescale, set by the product $(1 + \lambda_A A_{\text{typ}})/\tau_F$; the $O(\delta A)$ correction vanishes at the operating point. By the same argument as in Pair 1, the gradients of the $F$-decay rate with respect to $\tau_F$ and $\lambda_A$ at the operating point are scalar multiples of one another, so the corresponding FIM contribution is rank one. The combination resolved is the inverse effective rate $1/\tau_F^{\text{eff}}$.

### A.4 The null space of the FIM

Combining the three rank-one defects of Sections A.1–A.3, the FIM of the pre-G1 model evaluated at the operating point has a null space of dimension three. The three null vectors point along

$$(\delta\kappa_B, \delta\varepsilon_A) \perp \nabla\kappa_B^{\text{eff}}, \quad (\delta\mu_F, \delta\mu_{FF}) \perp \nabla\mu_F^{\text{eff}}, \quad (\delta\tau_F, \delta\lambda_A) \perp \nabla(1/\tau_F^{\text{eff}}), \tag{46}$$

i.e. along perturbations of each pair that hold the strongly identified linear combination fixed. The total FIM rank is therefore $d - 3$, where $d$ is the full dimension of the parameter vector.

The remaining $d - 3$ identified directions include all observation-model parameters (which couple linearly to the latent state and are therefore generically identifiable from a one-day record under the sleep- and wake-gating structure of (8)–(11)), the slow Banister parameters $\tau_B$ and $\kappa_F$ (which appear linearly and not in collinear pairs), and the diffusion scales $\sigma_B, \sigma_F, \sigma_A$ (which are identifiable from the empirical quadratic variation of the observation residuals).

### A.5 The G1 reparametrization is mathematically equivalent to the pre-G1 model

**Proposition 16** (G1 is mathematically equivalent to v2)**.** *The G1 drift* (3)–(5) *with the truth values of Table 1 is identically equal, as a function of* $(B, F, A, \Phi)$*, to the pre-G1 drift under the*

*relabelling*

$$\kappa_B^{\text{old}} = \kappa_B^{\text{new}}/(1+\varepsilon_A A_{\text{typ}}), \qquad \tau_F^{\text{old}} = \tau_F^{\text{new}}\,(1+\lambda_A A_{\text{typ}}),$$
$$\mu_F^{\text{old}} = \mu_F^{\text{new}} - 2F_{\text{typ}}\mu_{FF}, \qquad \mu_0^{\text{old}} = \mu_0^{\text{new}} - \mu_{FF}F_{\text{typ}}^2,$$

*with* $\varepsilon_A, \lambda_A, \mu_{FF}, \mu_B, \eta, \tau_B, \kappa_F$ *all unchanged.*

*Proof.* Substitute the relabelling into the pre-G1 drift terms (3)–(5) (in their original v2 form) and collect:

*B-equation.* The pre-G1 $B$-drift is $\kappa_B^{\text{old}}(1+\varepsilon_A A)\Phi - B/\tau_B$. Substituting $\kappa_B^{\text{old}} = \kappa_B^{\text{new}}/(1+\varepsilon_A A_{\text{typ}})$ gives $\kappa_B^{\text{new}} \cdot (1+\varepsilon_A A)/(1+\varepsilon_A A_{\text{typ}}) \cdot \Phi - B/\tau_B$, which is exactly the G1 $B$-drift (3).

*F-equation.* The pre-G1 $F$-drift is $\kappa_F\Phi - (1+\lambda_A A)F/\tau_F^{\text{old}}$. Substituting $\tau_F^{\text{old}} = \tau_F^{\text{new}}(1+\lambda_A A_{\text{typ}})$ gives $\kappa_F\Phi - (1+\lambda_A A)/(1+\lambda_A A_{\text{typ}}) \cdot F/\tau_F^{\text{new}}$, which is exactly the G1 $F$-drift (4).

*A-equation.* The growth rate $\mu(B,F)$ becomes, under the relabelling,

$$\begin{aligned}
\mu_0^{\text{old}} + \mu_B B - \mu_F^{\text{old}} F - \mu_{FF}F^2 &= (\mu_0^{\text{new}} - \mu_{FF}F_{\text{typ}}^2) + \mu_B B - (\mu_F^{\text{new}} - 2F_{\text{typ}}\mu_{FF})F - \mu_{FF}F^2 \\
&= \mu_0^{\text{new}} + \mu_B B - \mu_F^{\text{new}} F - \mu_{FF}(F^2 - 2F_{\text{typ}}F + F_{\text{typ}}^2) \\
&= \mu_0^{\text{new}} + \mu_B B - \mu_F^{\text{new}} F - \mu_{FF}(F - F_{\text{typ}})^2,
\end{aligned}$$

which is exactly the G1 growth rate (2). The $A$-drift is then $\mu(B,F)A - \eta A^3$, identical in both parametrizations. □ □

The numerical truth values of Table 1 satisfy the above relabelling exactly, so the unit test `tests/test_g1_reparam.py` – which evaluates the two drift formulas at 50 random states – agrees to floating-point precision. The reparametrization is purely a coordinate change in parameter space; the dynamics, the observation model, and the joint likelihood $p(y_{1:T} \mid \theta)$ are unchanged.

# B Stability of the FSA-v2 SDE: derivations

This appendix supplies the derivations underlying the stability analysis of Section 4. We compute the drift Jacobian that establishes Proposition 5 (Section B.1), set out the bifurcation locus explicitly (Section B.2), and prove the Foster–Lyapunov drift inequality of Theorem 7 (Section B.3).

## B.1 The drift Jacobian at the $A = 0$ branch

Write the deterministic drift vector field as $f(B,F,A;\Phi_c) = (f_B, f_F, f_A)^\top$ with

$$\begin{aligned}
f_B(B,F,A) &= \kappa_B\, g_B(A)\,\Phi_c - B/\tau_B, \\
f_F(B,F,A) &= \kappa_F\,\Phi_c - g_F(A)\,F/\tau_F, \\
f_A(B,F,A) &= \mu(B,F)\,A - \eta\,A^3,
\end{aligned}$$

where $g_B(A) = (1+\varepsilon_A A)/(1+\varepsilon_A A_{\text{typ}})$ and $g_F(A) = (1+\lambda_A A)/(1+\lambda_A A_{\text{typ}})$ are the residual factors. Their derivatives at $A = 0$ are $g_B'(0) = \varepsilon_A/(1+\varepsilon_A A_{\text{typ}})$ and $g_F'(0) = \lambda_A/(1+\lambda_A A_{\text{typ}})$.

The Jacobian $J = \partial f/\partial(B,F,A)$ at $(B_0^*, F_0^*, 0)$ has entries

- Row $B$: $\partial f_B/\partial B = -1/\tau_B$, $\partial f_B/\partial F = 0$, $\partial f_B/\partial A = \kappa_B g_B'(0)\Phi_c = \kappa_B\varepsilon_A\Phi_c/(1+\varepsilon_A A_{\text{typ}})$.
- Row $F$: $\partial f_F/\partial B = 0$, $\partial f_F/\partial F = -g_F(0)/\tau_F = -1/[\tau_F(1+\lambda_A A_{\text{typ}})]$, $\partial f_F/\partial A = -g_F'(0)F_0^*/\tau_F = -\lambda_A F_0^*/[\tau_F(1+\lambda_A A_{\text{typ}})]$.
- Row $A$: $\partial f_A/\partial B = \mu_B \cdot A = 0$ (at $A = 0$), $\partial f_A/\partial F = (-\mu_F - 2\mu_{FF}(F - F_{\text{typ}})) \cdot A = 0$ (at $A = 0$), $\partial f_A/\partial A = \mu(B,F) - 3\eta A^2 = \mu(B_0^*, F_0^*)$ (at $A = 0$).

In matrix form,

$$J(B_0^*, F_0^*, 0) \;=\; \begin{pmatrix} -1/\tau_B & 0 & \dfrac{\kappa_B \varepsilon_A \Phi_c}{1+\varepsilon_A A_{\text{typ}}} \\ 0 & -\dfrac{1}{\tau_F(1+\lambda_A A_{\text{typ}})} & -\dfrac{\lambda_A F_0^*}{\tau_F(1+\lambda_A A_{\text{typ}})} \\ 0 & 0 & \mu(B_0^*, F_0^*) \end{pmatrix}. \tag{47}$$

This is upper triangular, so its eigenvalues are the diagonal entries:

$$\lambda_1 = -1/\tau_B, \qquad \lambda_2 = -1/[\tau_F(1+\lambda_A A_{\text{typ}})], \qquad \lambda_3 = \mu(B_0^*, F_0^*). \tag{48}$$

The first two are unconditionally negative (with $\tau_B, \tau_F > 0$ and $\lambda_A, A_{\text{typ}} \geq 0$); the third has the sign of the Stuart–Landau growth rate. By Lyapunov's first method, the $A = 0$ equilibrium is locally asymptotically stable iff all three eigenvalues are negative, i.e. iff $\mu(B_0^*, F_0^*) < 0$. This proves Proposition 5. □

## B.2 The bifurcation locus

The bifurcation locus $\{\Phi_c > 0 : \mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) = 0\}$ is the level set on which $\lambda_3$ in (48) vanishes. Substituting $B_0^*(\Phi_c) = \tilde{b}\,\Phi_c$ and $F_0^*(\Phi_c) = \tilde{f}\,\Phi_c$ with $\tilde{b} = \kappa_B \tau_B/(1+\varepsilon_A A_{\text{typ}})$ and $\tilde{f} = \kappa_F \tau_F (1+\lambda_A A_{\text{typ}})$ into the formula for $\mu$,

$$\mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) \;=\; \mu_0 + \mu_B \tilde{b}\,\Phi_c - \mu_F \tilde{f}\,\Phi_c - \mu_{FF}(\tilde{f}\Phi_c - F_{\text{typ}})^2. \tag{49}$$

This is a quadratic in $\Phi_c$ once expanded, with leading coefficient $-\mu_{FF}\tilde{f}^2 < 0$. At the truth values of Table 1,

$$\tilde{b} \;=\; \frac{0.01248 \cdot 42}{1.04} \;\approx\; 0.504, \qquad \tilde{f} \;=\; 0.030 \cdot 6.364 \cdot 1.1 \;\approx\; 0.210,$$

giving $\mu(\Phi_c) = 0.036 + 0.0966\,\Phi_c - 0.40(0.210\,\Phi_c - 0.20)^2$. Solving $\mu(\Phi_c) = 0$ yields a single positive root $\Phi_c^{\text{crit}} \approx 7.5$ TRIMP-units per day.

For $0 < \Phi_c < \Phi_c^{\text{crit}}$ we have $\mu > 0$ and the $A = 0$ branch is unstable; for $\Phi_c > \Phi_c^{\text{crit}}$ we have $\mu < 0$ and the $A = 0$ branch is stable. At $\Phi_c = \Phi_c^{\text{crit}}$ the system undergoes a Hopf-like (super- or sub-critical depending on the sign of higher-order corrections we have not computed) bifurcation in the $A$-direction.

## B.3 Foster–Lyapunov drift inequality (proof of Theorem 7)

We now prove that the quadratic Lyapunov function $V(B, F, A) = \frac{1}{2}(aB^2 + bF^2 + cA^2)$ of (15) satisfies the Foster–Lyapunov drift inequality (14) on the operating domain $\mathcal{X} = [0,1] \times [0,\infty)^2$, with rate $\alpha = \min(2/\tau_B,\, 1/[\tau_F(1+\lambda_A A_{\text{typ}})])$ and a finite constant $C > 0$.

Let $\mathcal{L}$ denote the infinitesimal generator of the SDE: for any $V \in C^2(\mathcal{X})$,

$$\mathcal{L}V(x) \;=\; \sum_{i \in \{B,F,A\}} b_i(x)\,\partial_i V(x) \;+\; \tfrac{1}{2}\sum_i a_i(x)\,\partial_{ii}^2 V(x), \tag{50}$$

with $b_i$ the $i$th drift component and $a_B(x) = \sigma_B^2 B(1-B)$, $a_F(x) = \sigma_F^2 F$, $a_A(x) = \sigma_A^2 A$. We bound the three components separately. Throughout, $K_\bullet$ denotes a finite constant depending only on the truth parameters, $\Phi_{\max}$, and the choice of $a, b, c$.

**$B$-component.** Using $B \in [0,1]$ and the fact that $g_B(A) = (1+\varepsilon_A A)/(1+\varepsilon_A A_{\text{typ}})$ is bounded above on the operating domain by some $g_{B,\max} < \infty$ (depending on the practical upper bound on $A$ that we wish to enforce; on the operating domain $A \in [0, A_{\max}]$ with any chosen $A_{\max}$, set $g_{B,\max} = (1+\varepsilon_A A_{\max})/(1+\varepsilon_A A_{\text{typ}})$),

$$\begin{aligned} aB \cdot \big[\kappa_B g_B(A)\Phi - B/\tau_B\big] + \tfrac{a}{2}\sigma_B^2 B(1-B) &\le a\,\kappa_B\, g_{B,\max}\,\Phi_{\max}\, B - (a/\tau_B)\, B^2 + \tfrac{a}{8}\sigma_B^2 \\ &\le a\,\kappa_B\, g_{B,\max}\,\Phi_{\max} + \tfrac{a}{8}\sigma_B^2 - (a/\tau_B)\, B^2, \end{aligned}$$

where the second line uses $B \le 1$ and the bound $B(1-B) \le 1/4$ on $[0,1]$. Define $K_B := a\kappa_B g_{B,\max}\Phi_{\max} + a\sigma_B^2/8$.

**$F$-component.** Using $g_F(A) \ge g_F(0) = 1/(1+\lambda_A A_{\text{typ}})$ for $A \ge 0$, and writing $\tau_F^{\text{old}} := \tau_F(1+\lambda_A A_{\text{typ}})$,

$$\begin{aligned} bF \cdot \big[\kappa_F\Phi - g_F(A)F/\tau_F\big] + \tfrac{b}{2}\sigma_F^2 F &\le b\,\kappa_F\,\Phi_{\max}\, F - bF^2/\tau_F^{\text{old}} + \tfrac{b}{2}\sigma_F^2 F \\ &= bF\,(\kappa_F\Phi_{\max} + \sigma_F^2/2) - bF^2/\tau_F^{\text{old}}. \end{aligned}$$

Apply Young's inequality $xy \le x^2/(2\beta) + \beta y^2/2$ to the linear-in-$F$ term, with $\beta = \tau_F^{\text{old}}$ and $y = (\kappa_F\Phi_{\max} + \sigma_F^2/2)$:

$$bF(\kappa_F\Phi_{\max} + \sigma_F^2/2) \;\le\; bF^2/(2\tau_F^{\text{old}}) + (b\tau_F^{\text{old}}/2)(\kappa_F\Phi_{\max} + \sigma_F^2/2)^2.$$

Combined,

$$bF \cdot \big[\kappa_F\Phi - g_F(A)F/\tau_F\big] + \tfrac{b}{2}\sigma_F^2 F \;\le\; -(b/(2\tau_F^{\text{old}}))F^2 + K_F,$$

with $K_F := (b\tau_F^{\text{old}}/2)(\kappa_F\Phi_{\max} + \sigma_F^2/2)^2$.

**$A$-component.** The growth rate satisfies $\mu(B,F) \le \mu_{\max} := \mu_0 + \mu_B$ for $B \in [0,1]$ and $F \ge 0$, since the contributions $-\mu_F F$ and $-\mu_{FF}(F - F_{\text{typ}})^2$ are non-positive. Therefore

$$cA\big[\mu(B,F)\,A - \eta A^3\big] + \tfrac{c}{2}\sigma_A^2 A \;\le\; -c\eta A^4 + c\mu_{\max}A^2 + \tfrac{c}{2}\sigma_A^2 A.$$

Two applications of Young's inequality remove the unwanted $A^2$ and linear-$A$ terms:

*(i)* For the $A^4$-vs-$A^2$ pair, complete the square with $x = A^2$:

$$-c\eta A^4 + c\mu_{\max}A^2 = -c\eta\big(A^2 - \mu_{\max}/(2\eta)\big)^2 + c\mu_{\max}^2/(4\eta) \;\le\; -\tfrac{c\eta}{2}A^4 + \tfrac{c\mu_{\max}^2}{2\eta},$$

where we discarded half of the negative quartic contribution to absorb the maximum of the non-negative quadratic.

*(ii)* For the linear $A$-term against the diffusion: by the elementary $A \le (\eta/(2\sigma_A^2))A^4 + \sigma_A^2/(2\eta)$ (setting $\alpha = \eta/(2\sigma_A^2)$ in $x \le \alpha x^4 + 1/(4\alpha)$ – proved by completing the square on $x \mapsto \alpha x^4 - x$ and verifying the minimum is $-1/(4\alpha)$, so $x - \alpha x^4 \le 1/(4\alpha)$, i.e. $x \le \alpha x^4 + 1/(4\alpha)$), and we then absorb the $A^4$ piece into a small fraction of the $-\frac{c\eta}{2}A^4$ term:

$$\tfrac{c}{2}\sigma_A^2 A \;\le\; \tfrac{c}{2}\sigma_A^2 \cdot \big(\eta/(2\sigma_A^2)A^4 + \sigma_A^2/(2\eta)\big) \;=\; \tfrac{c\eta}{4}A^4 + \tfrac{c\sigma_A^4}{4\eta}.$$

Wait – that produces the wrong sign on $A^4$. We instead want the linear-$A$ term to sit on top of the cubic damping; redo with the opposite split. Use $A \le 1 + A^2$ (a strict bound for $A \ge 0$ verified by completing the square on $1 + A^2 - A$); then

$$\tfrac{c}{2}\sigma_A^2 A \;\le\; \tfrac{c\sigma_A^2}{2}(1 + A^2) \;=\; \tfrac{c\sigma_A^2}{2} + \tfrac{c\sigma_A^2}{2}A^2,$$

and the residual $A^2$ is then absorbed via the same complete-the-square argument as in step (i), but with $\mu_{\max}$ replaced by $\sigma_A^2/2$. Combining all three pieces,

$$cA[\mu(B,F)A - \eta A^3] + \tfrac{c}{2}\sigma_A^2 A \;\le\; -\tfrac{c\eta}{4}A^4 + \tfrac{c\mu_{\max}^2}{2\eta} + \tfrac{c\sigma_A^2}{2} + \tfrac{c\sigma_A^4}{8\eta} \;=:\; -\tfrac{c\eta}{4}A^4 + K_A.$$

**Combining the three components.** Adding the three bounds,

$$\mathcal{L}V(x) \;\le\; -(a/\tau_B)\,B^2 - \big(b/(2\tau_F^{\text{old}})\big)F^2 - (c\eta/4)\,A^4 + (K_B + K_F + K_A).$$

The first two negative terms already match $-\alpha V$ with $\alpha = \min(2/\tau_B,\, 1/\tau_F^{\text{old}})$. For the third we use the elementary inequality

$$-(c\eta/4)\,A^4 \;\le\; -(c\alpha/2)\,A^2 + c\alpha^2/(4\eta),$$

which follows by completing the square in $A^2$ on the parabola $(c\eta/4)x^2 - (c\alpha/2)x$. Plugging this in,

$$\mathcal{L}V(x) \;\le\; -\tfrac{\alpha}{2}\big(aB^2 + bF^2 + cA^2\big) + \big(K_B + K_F + K_A + c\alpha^2/(4\eta)\big) \;=\; -\alpha\,V(x) + C,$$

where $C := K_B + K_F + K_A + c\alpha^2/(4\eta)$ is finite. This is the required drift inequality (14), completing the proof of Theorem 7. □

The exponential ergodicity statement of Theorem 8 then follows from the standard Foster–Lyapunov ⇒ V-uniform ergodicity machinery in Khasminskii (1980) or, in discrete-time language transferable to continuous time via the resolvent, Meyn–Tweedie (1993, Theorem 6.1). The constants $K_0$ and $\rho$ in (16) can be made explicit at the cost of further notation; we do not pursue this here.

# C Numerical stability of the SDE solver: derivations

This appendix supplies the algebra underlying the stability claims of Section 5. We derive the mean-square stability condition for explicit Euler–Maruyama (Section C.1), apply it to FSA-v2 component by component (Section C.2), analyse the role of the inner substepping for the cubic Stuart–Landau term (Section C.3), and discuss the interaction of the discretised CIR diffusion with the Feller boundary condition (Section C.4).

## C.1 The linear test SDE and explicit Euler–Maruyama

Consider the linear scalar test SDE

$$\mathrm{d}X_t = a\,X_t\,\mathrm{d}t + b\,X_t\,\mathrm{d}W_t, \qquad a, b \in \mathbb{R}, \quad X_0 \in \mathbb{R}, \tag{51}$$

whose continuous-time solution is the geometric Brownian motion $X_t = X_0 \exp\big((a - b^2/2)t + bW_t\big)$. Mean-square stability of (51) – i.e. $\mathbb{E}[X_t^2] \to 0$ as $t \to \infty$ – holds iff $2a + b^2 < 0$ (Higham, 2000).

The explicit Euler–Maruyama discretisation of (51) with step $h$ is

$$X_{n+1} \;=\; X_n + a\,X_n\,h + b\,X_n\,\sqrt{h}\,\xi_n, \qquad \xi_n \sim \mathcal{N}(0,1) \text{ i.i.d.} \tag{52}$$

Squaring and taking expectations,

$$\begin{aligned}\mathbb{E}[X_{n+1}^2] &= \mathbb{E}\big[(1+ah)^2 X_n^2 + 2(1+ah)b\sqrt{h}X_n^2\xi_n + b^2 h X_n^2 \xi_n^2\big] \\ &= \big((1+ah)^2 + b^2 h\big)\,\mathbb{E}[X_n^2],\end{aligned}$$

so $\mathbb{E}[X_n^2]$ remains bounded iff

$$(1+ah)^2 + b^2 h \;\le\; 1. \tag{53}$$

This is the inequality (18) of the main text. Expanding $(1+ah)^2 = 1 + 2ah + a^2h^2$ and rearranging, (53) is equivalent to

$$2a + b^2 + a^2\,h \;\le\; 0, \tag{54}$$

which collapses to the continuous-time condition $2a + b^2 < 0$ as $h \to 0$. For $a < 0$, the discretisation imposes the additional constraint $h \le -(2a + b^2)/a^2$, i.e. $h \le 2|a|/(a^2)$ when $b = 0$.

## C.2 Component-by-component bounds for FSA-v2

We linearise the FSA-v2 SDE component by component around the $A = 0$ deterministic equilibrium $(B_0^*, F_0^*, 0)$ of Section 4.1. Write $\delta B = B - B_0^*$ etc. The linearised $B$-equation reads

$$\mathrm{d}(\delta B) \;=\; -(\delta B/\tau_B)\,\mathrm{d}t + \sigma_B\sqrt{B_0^*(1-B_0^*)}\,\mathrm{d}W_B + (\text{coupling terms}).$$

Treating the coupling terms as forcing, the homogeneous part is the test SDE (51) with $a = -1/\tau_B$ and effective $b_{\text{eff}}^2 = \sigma_B^2 B_0^*(1-B_0^*)/(\delta B)^2$. Since $\delta B$ ranges over the order-unity scale on which we want the filter to operate, $b_{\text{eff}}^2$ is bounded above by $\sigma_B^2/4$ (the maximum of $B(1-B)$ on $[0,1]$). Plugging into (53),

$$(1-h/\tau_B)^2 + (\sigma_B^2/4)\,h \;\le\; 1 \iff h \;\le\; \frac{2/\tau_B - \sigma_B^2/4}{1/\tau_B^2} \;\approx\; 2\tau_B \;=\; 84\,\mathrm{d},$$

where the diffusion contribution $\sigma_B^2/4 \approx 2.5\times 10^{-5}$ is negligible compared to $2/\tau_B \approx 0.048\,\mathrm{d}^{-1}$.

The linearised $F$-equation has $a = -1/\tau_F^{\text{old}}$ and $b_{\text{eff}}^2 = \sigma_F^2 F_0^*/(\delta F)^2$, bounded similarly. The same calculation gives $h \le 2\tau_F^{\text{old}} = 14\,\mathrm{d}$.

The $A$-equation around $A = 0$ has $a = \mu(B_0^*, F_0^*)$, which can be *positive* below the bifurcation locus. In that case the $A = 0$ branch of the deterministic skeleton is unstable (Proposition 5) and the natural linearisation point is the oscillatory branch instead, where $a$ becomes $-2\eta(A^*)^2$. At the limit-cycle amplitude $A^* = \sqrt{\mu/\eta} \approx 0.81$ (using $\mu \approx 0.13$, $\eta = 0.20$), we have $a = -2\cdot 0.20\cdot 0.66 \approx -0.27\,\mathrm{d}^{-1}$, and $b_{\text{eff}}^2 = \sigma_A^2 A^*/(\delta A)^2 \le \sigma_A^2/(A^*)$, which is also negligible. Plugging in, $h \le 2/0.27 \approx 7.4\,\mathrm{d}$.

The binding component is therefore the $A$-mode at the limit cycle, giving the practical stability ceiling of roughly seven days. The current outer step $\Delta t = 1/96\,\mathrm{d}$ is below this ceiling by a factor of 700 – the quoted "three orders of magnitude" of the main text.

## C.3 The role of the inner substepping

The inner-loop substepping of (17) runs $n_{\text{sub}} = 4$ deterministic Euler steps of size $\Delta t/n_{\text{sub}} = \Delta t/4$ before applying the single Wiener increment. We now check that the substepping is neither necessary for stability (which it isn't, given the factor-700 margin above) nor harmful for it.

A pure deterministic Euler step on the cubic Stuart–Landau ODE $\dot{A} = \mu A - \eta A^3$ around the limit-cycle amplitude $A^* = \sqrt{\mu/\eta}$ has linearised rate $a_{\text{det}} = -2\eta(A^*)^2 = -2\mu \approx -0.27\,\mathrm{d}^{-1}$. The deterministic Euler stability condition $|1 + a_{\text{det}}h| \le 1$ becomes $h \le 2/|a_{\text{det}}| \approx 7.4\,\mathrm{d}$, identical to the bound computed above. For the inner step $h_{\text{sub}} = \Delta t/4$ to violate this bound would require $\Delta t \ge 30\,\mathrm{d}$, far beyond any contemplated step.

The substepping is therefore a *accuracy* refinement rather than a stability device: at large excursions $A \gg A^*$ where the cubic damping becomes very fast, the inner loop reduces the deterministic-step error from $O(\Delta t)$ to $O(\Delta t/n_{\text{sub}})$ at the cost of evaluating the drift $n_{\text{sub}}$ times per outer step. The Wiener increment is applied once at the outer-step boundary, so the stochastic discretisation continues to inherit the strong order-1/2 convergence of explicit Euler–Maruyama.

## C.4 The CIR diffusion and the Feller condition

The $F$- and $A$-components of the SDE have square-root (CIR-type) diffusions $\sigma_F\sqrt{F}\,\mathrm{d}W_F$ and $\sigma_A\sqrt{A}\,\mathrm{d}W_A$. A continuous-time CIR process $\mathrm{d}X = (\alpha - \beta X)\,\mathrm{d}t + \nu\sqrt{X}\,\mathrm{d}W$ remains strictly positive almost surely if and only if it satisfies the *Feller condition*

$$2\alpha \;>\; \nu^2. \tag{55}$$

For the $F$-component at constant $\Phi = 1$, $\alpha = \kappa_F\cdot 1 = 0.030$ and $\nu^2 = \sigma_F^2 = 1.44\times 10^{-4}$, so $2\alpha/\nu^2 \approx 4.2\times 10^2 \gg 1$ – comfortably satisfied. The $A$-component does not directly fit the CIR

template (the drift is multiplicative, $\mu A - \eta A^3$, not affine), but the analogous condition near $A = A^*_{\text{osc}}$ involves the ratio $2\mu A^*_{\text{osc}}/\sigma_A^2 = 2\mu^{3/2}/(\eta^{1/2}\sigma_A^2)$, also large for the truth values.

The discretised explicit Euler–Maruyama scheme can nevertheless produce negative values of $F$ or $A$ on rare excursions, even when (55) holds, because of the $\sqrt{\Delta t}\,\xi_n$ Wiener increment: a single $|\xi_n| \gg 1$ realisation can push the post-step state below zero. The reflection step in (17) catches these excursions and folds them back into the physiological domain. In practice the diffusion vanishing at the boundary ($\sigma_F\sqrt{F} \to 0$ as $F \to 0$, similarly for $A$) makes such excursions extraordinarily rare; the Feller-aware schemes of e.g. Higham, Mao and Stuart (2003) would handle them more cleanly but are not currently implemented.

### C.5 What an enlarged step would buy

Stage D and Stage E5 of Section 10 ran on the 15-minute outer step. The wall-clock cost of those runs was $\sim$55 minutes (Stage D, $T = 84$ days) and $\sim$41 minutes (Stage E5, $T = 14$ days, 27 rolling windows) on a single GPU. For a fixed simulation horizon $T$, the SDE solver cost scales linearly in the number of outer steps $T/\Delta t$. Doubling $\Delta t$ from 15 to 30 minutes halves that count and hence the cost. The full battery of horizon-sweep experiments (Section 11.4) at $T \in \{28, 42, 56, 84\}$ days would scale similarly, taking the long $T = 84$-day closed-loop run from $\sim$15 hours down to $\sim$7.5 hours at $\Delta t = 30\,\text{min}$.

The numerical-stability analysis of this appendix supports such an enlargement up to $\Delta t \approx 1\,\text{hour}$ without a solver upgrade, conditional on a one-off drift-parity verification against the 15-minute reference. Beyond one hour the $\Phi$-burst-resolution constraint of Section 5.5 starts to dominate, and beyond that the Lamperti / Milstein / implicit upgrades flagged in Remark 10 become the right route.